ACADEMIC REGULATIONS
COURSE STRUCTURE
AND
DETAILED SYLLABUS

R24

Bachelor of Computer Applications (Honours)

B.C.A/B.C.A (H)

Four Years U.G. Programme

(Applicable for the batches admitted from the A.Y. 2024-2025)

SWARNANDHRA

COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

SEETHARAMAPURAM, NARSAPUR-534 280, W.G.DT., A.P.

ACADEMIC REGULATIONS R24

ACADEMIC REGULATIONS R24 FOR BCA / B.C.A.,(Hons.) PROGRAMME

Applicable for the students of B.C.A./B.C.A.,(Hons.)(Regular) Programme admitted from the Academic Year 2024-25 and onwards

The **B.C.A./B.C.A.,(Hons.)**Degree shall be conferred on candidates who are admitted to the program and who fulfil all the requirements for the award of the Degree.

1.0 ELIGIBILITY FOR ADMISSIONS

Admission to the above program shall be made subject to eligibility, qualification and specialization as prescribed by the University from time to time.

Admissions shall be made on the basis of merit rank obtained by the candidates at entrance examination concerned or the qualifying Entrance Test conducted by the University or on the basis of any other order of merit as approved by the University, subject to reservations as laid down by the Govt. of Andhra Pradesh from time to time.

2.0 AWARD OF B.C.A. DEGREE

2.1 A student shall be declared eligible for the award of the B.C.A., Degree, if he pursues a course of study and completes it successfully in not less than THREE academic years and not more than SIX academic years

2.2 The student shall register for all 120 credits and secure all the 120 credits.

2.3 The minimum instruction days in each semester are 90.

2.4 A Student, who fails to fulfil all the academic requirements for the award of the degree within SIX academic years from the year of their admission, shall forfeit his seat in B.C.A., course

2.5 **After successful completion of First three-year academic requirements a student can opt to REGISTER for B.C.A. (Hons) Degree by completing 40 more credits in the FOURTH YEAR.**

3.0 ATTENDANCE REQUIREMENTS:

i) A student shall be eligible to appear for the University external examinations if he/she acquires a minimum of 75% of attendance in aggregate of all the subjects. Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester may be granted by the College Academic Committee.

ii) Shortage of Attendance below 65% in aggregate shall in NO CASE be condoned.

iii) A stipulated fee of Rs 500/- shall be payable towards condonation of shortage of attendance to the University.

iv) Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examination of that class and their registration shall stand cancelled.

v) A student will not be promoted to the next semester unless he satisfies the attendance requirements of the present semester. They may seek readmission for that semester from the date of commencement of class work.

vi) If any candidate fulfils the attendance requirement in the present semester, he shall not be eligible for readmission into the same class.

vii) If the learning is carried out in blended mode (both offline & online), then the total attendance of the student shall be calculated considering the offline and online attendance of the student.

a. Promotion Rules:

The following academic requirements must be satisfied in addition to the attendance requirements mentioned in the above.

i) A student shall be promoted from first year to second year if he/she fulfils the minimum attendance requirement as per university norms.

ii) A student will be promoted from II to III year if he/she fulfils the academic requirement of securing 40% of the credits (any ***decimal*** fraction should be ***rounded off*** to **higher** digit) up to in the subjects that have been studied up to III or IV semester.

iii) A student shall be promoted from III year to IV year *(for B.C.A. (Hons))* if he/she fulfils the academic requirements in the subjects that have been studied up to V or VI semester.

And in case a student is detained for want of credits for a particular academic year by ii) above, the student may make up the credits through supplementary examinations and only after securing the required credits he/she shall be permitted to join in the V semester.

When a student is detained due to lack of credits/shortage of attendance he/she may be re-admitted when the semester is offered after fulfilment of academic regulations.

4.0 EVALUATION

The performance of a student in each semester shall be evaluated subject wise with a maximum of 100 marks for theory and 100 marks for practical subject. Summer Internships, Minor Project with Seminar shall be evaluated for 50 marks, Research Project / Dissertation work in VIII semester (BCA (Hons.)) shall be evaluated for 200 marks, mandatory courses with no credits shall be evaluated for 30 mid semester marks.

A student has to secure not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid semester and end examination marks taken together for the theory, practical or project etc. In case of a mandatory course, he/she should secure 40% of the total marks.

Theory Courses

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	**100**

i) For theory subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End-Examination.

ii) For practical subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End- Examination.

iii) If any subject is having both theory and practical components, they will be evaluated separately as theory subject and practical subject. However, they will be given same subject code with an extension of 'T' for theory subject and 'L' for practical subject.

a) Continuous Internal Evaluation

i) For theory subjects, during the semester, there shall be two midterm examinations. Each midterm examination shall be evaluated for 30 marks of which 10 marks for objective paper (20 minutes' duration), 15 marks for subjective paper (90 minutes' duration) and 5 marks for assignment. ii) Objective paper shall contain for 05 short answer questions with 2 marks each or maximum of 20 bits for 10 marks. Subjective paper shall contain 3 either or type questions (totally six questions from 1 to 6) of which student has to answer one from each either-or type of questions. Each question carries 10 marks. The marks obtained in the subjective paper are condensed to 15 marks.

Note:

- The objective paper shall be prepared in line with the quality of competitive examinations questions.

- The subjective paper shall contain 3 either or type questions of equal weightage of 10 marks. Any fraction shall be rounded off to the next higher mark.
- The objective paper shall be conducted by the respective institution on the day of subjective paper test
- Assignments shall be in the form of problems, mini projects, design problems, slip tests, quizzes etc., depending on the course content. It should be continuous assessment throughout the semester and the average marks shall be considered.

iii) If the student is absent for the mid semester examination, no re-exam shall be conducted and mid semester marks for that examination shall be considered as zero.

iv) First midterm examination shall be conducted for I, II units of syllabus with one either or type question from each unit and third either or type question from both the units. The second midterm examination shall be conducted for III, IV and V units with one either or type question from each unit.

v) Final mid semester marks shall be arrived at by considering the marks secured by the student in both the mid examinations with 80% weightage given to the better mid exam and 20% to the other.

For Example:

Marks obtained in first mid: 25
Marks obtained in second mid: 20
Final mid semester Marks: (25x0.8) + (20x0.2) = 24

If the student is absent for any one midterm examination, the final mid semester marks shall be arrived at by considering 80% weightage to the marks secured by the student in the appeared examination and zero to the other. For Example:

Marks obtained in first mid: Absent
Marks obtained in second mid: 25
Final mid semester Marks: (25x0.8) + (0x0.2) =20

b) End Examination Evaluation:

End examination of theory subjects shall have the following pattern:

The **end semester** examination will be conducted for 70 marks which covers full syllabus. In end examination pattern, **Part–A** consists of five short questions from all units for 10 marks. **Part – B** has **5 questions** with internal choice from each unit and valued for 60marks.

Practical Courses

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	**100**

a) For practical courses, there shall be a continuous evaluation during the semester for 30 sessional marks and end examination shall be for 70 marks.

b) Day-to-day work in the laboratory shall be evaluated for 15 marks by the concerned laboratory teacher based on the record/viva and 15 marks for the internal test.

c) The end examination shall be evaluated for 70 marks, conducted by the concerned laboratory teacher and outside external subject expert.

   i. Procedure: 20 marks
   ii. Experimental work & Results: 30 marks

iii. Viva voce: 20 marks.

d) There shall be no external examination for mandatory courses with zero credits. However, attendance shall be considered while calculating aggregate attendance and student shall be declared to have passed the mandatory course only when he/she secures 40% or more in the internal examinations. In case, the student fails, a re-examination shall be conducted for failed candidates for 30 marks satisfying the conditions mentioned in item 1 & 2 of the regulations.

e) The laboratory records and mid semester test papers shall be preserved for a minimum of 3 years in the respective institutions as per the University norms and shall be produced to the Committees of the University as and when the same are asked for.

4.1 There shall be an internship, one need to complete during year break (*i.e., IV sem to V Sem*) and will be evaluated for 50 marks internally at the end of V Semester by the departmental committee. A candidate has to secure a minimum 50% of marks to be declared successful (*for 2 credits*).

4.2 There shall be a mini project, during VI Sem and will be evaluated for 50 marks internally at the end of VI Semester by the departmental committee. A candidate has to secure a minimum 50% of marks to be declared successful (*for 2 credits*).

4.3 There shall be a **Research project/ Dissertation** for B.C.A. (Hons) students, during VIII Sem. (*for 12 credits*).

4.4 For audit courses, attendance is mandatory and a candidate has to get a satisfactory report for successful completion

4.5 A candidate shall be allowed to submit the Research Project / Dissertation work only after fulfilling the attendance requirements of all the semesters. The viva-voce examination shall be conducted at the end of the course work for **B.C.A. (Hons.)** (VIII semester).

MOOCs Courses:

1. Massive Open Online Courses (MOOCs):

A Student has to pursue and complete one course compulsorily through MOOCs approved by the University. A student can pursue courses other than core through MOOCs and it is mandatory to complete one course successfully through MOOCs for awarding the degree. A student is not permitted to register and pursue core courses through MOOCs.

A student shall register for the course (Minimum of either 8 weeks or 12 weeks) offered through MOOCs with the approval of Head of the Department. The Head of the Department shall appoint one mentor to monitor the student's progression. The student needs to earn a certificate by passing the exam. The student shall be awarded the credits assigned in the curriculum only by submission of the certificate. Examination fee, if any, will be borne by the student.

Students who have qualified in the proctored examinations conducted through MOOCs platform can apply for credit transfer as specified and are exempted from appearing internal as well as external examination (for the specified equivalent credit course only) conducted by the university.
Necessary amendments in rules and regulations regarding adoption of MOOC courses would be proposed from time to time.

2. Credit Transfer Policy

Adoption of MOOCs is mandatory, to enable Blended model of teaching-learning as also envisaged in the NEP 2020. As per University Grants Commission (Credit Framework for Online Learning Courses through SWAYAM) Regulation, 2016, the University shall allow up to a maximum of 20% of the total courses being offered in a particular programme i.e., maximum of 32 credits through MOOCs platform.

i) The University shall offer credit mobility for MOOCs and give the equivalent credit weightage to the students for the credits earned through online learning courses.

ii) Student registration for the MOOCs shall be only through the respective department of the institution, it is mandatory for the student to share necessary information with the department. ***Student can register ONE Semester in advance and obtain required credits, i.e. e.g,, V Sem courses can be register while studying IV Sem itself and can obtain credits***.

iii) The concerned department shall identify the courses permitted for credit transfer.

iv) The University/institution shall notify at the beginning of semester the list of the online learning courses eligible for credit transfer.

v) The institution shall designate a faculty member as a Mentor for each course to guide the students from registration till completion of the credit course.

vi) The university shall ensure no overlap of MOOC exams with that of the university examination schedule. In case of delay in results, the university will reissue the marks sheet for such students. vii) Student pursuing courses under MOOCs shall acquire the required credits only after successful completion of the course and submitting a certificate issued by the competent authority along with the percentage of marks and grades. viii) The institution shall submit the following to the examination section of the university:

a) List of students who have passed MOOC courses in the current semester along with the certificate of completion.

b) Undertaking form filled by the students for credit transfer. ix) The universities shall resolve any issues that may arise in the implementation of this policy from time to time and shall review its credit transfer policy in the light of periodic changes brought by UGC, SWAYAM, NPTEL and state government.

Note: Students shall be permitted to register for MOOCs offered through online platforms approved by the University from time to time.

5. Grading:

1. As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades and corresponding percentage of marks shall be followed:

After each course is evaluated for 100 marks, the marks obtained in each course will be converted to a corresponding letter grade as given below, depending on the range in which the marks obtained by the student fall.

Structure of Grading of Academic Performance

Range in which the marks in the subject fall	Grade	Grade points Assigned
90 & above	S (Superior)	10
80 to 89	A(Excellent)	9
70 to 79	B(Very Good)	8
60 to 69	C(Good)	7
50 to 59	D(Average)	6
40 to 49	E(Pass)	5
< 40	F (Fail)	0
Absent	Ab(Absent)	0

i) A student obtaining Grade 'F' or Grade 'Ab' in a subject shall be considered failed and will be required to reappear for that subject when it is offered the next supplementary examination.

ii) For non-credit audit courses, "Satisfactory" or "Unsatisfactory" shall be indicated instead of the letter grade and this will not be counted forth ecomputation of SGPA/CGPA/Percentage.

Computation of Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA):
The Semester Grade Point Average (SGPA)is the ratio of sum of the product of the number of credits with the grade points scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.,

$$SGPA= \Sigma(Ci \times Gi)/\Sigma Ci$$

where, Ci is the number of credits of the ith subject and Gi is the grade points cored by the student in the ith course.

The Cumulative Grade Point Average (CGPA) will be computed in the same manner considering all the courses undergone by a student overall the semesters of a program, i.e.,

$$CGPA= \Sigma(Ci \times Si)/\Sigma Ci$$

where "Si" is the SGPA of the ith semester and Ciis the total number of credits up to that semester.

Both SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts. While computing the SGPA the subjects in which the student is awarded Zero grade points will also be included.

Grade Point: It is a numerical weight allotted to each letter grade on a 10-point scale. Letter Grade: It is an index of the performance of students in a said course. Grades are denoted by the letters S, A, B, C, D ,E and F.

2. Award of Class:

Afterastudenthassatisfiedtherequirementsprescribedforthecompletionoftheprogram and is eligible for the award of the Degree, he/she shall be placed in one of the following four classes:

Class Awarded	CGPA Secured
First Class with Distinction	≥ 7.5
First Class	≥ 6.5 < 7.5
Second Class	≥ 5.5 < 6.5
Pass Class	≥ 5.0 < 5.5

CGPA to Percentage Conversion Formula – (CGPA – 0.5) x 10

3. With–holding of Results

If the candidate has any dues not paid to the university or if any case of indiscipline or malpractice is pending against him/her, the result of the candidate shall be withheld in such cases.

4. Multiple Entry / Exit Option

(a) Exit Policy:

The students can choose to exit the three/four-year programme at the end of first/second/third year, as the case may be.

- If a student wants to exit after obtaining 45 credits, he/she may be awarded with UG **Certificate in Computer Applications**

- If a student wants to exit after obtaining **85** credits, he/she may be awarded with UG **Diploma in Computer Applications**

(b) Entry Policy:

Modalities on multiple entry by the student into the B.C.A /B.C.A.(Hons) programme will be provided in due course of time.

Note: The Universities shall resolve any issues that may arise in the implementation of Multiple Entry and Exit policies from time to time and shall review the policies in the light of periodic changes brought by UGC, AICTE and State government.

5. Gap Year Concept:

Gap year concept for Student Entrepreneur in Residence is introduced and outstanding students who wish to pursue entrepreneurship / become entrepreneur are allowed to take a break of one year at any time after I year to pursue full-time entrepreneurship programme/to establish startups. *This period may be extended to two years at the most and these two years would not be counted for the time for the maximum time for graduation*. The principal of the respective college shall forward such proposals submitted by the students to the University. An evaluation committee constituted by the University shall evaluate the proposal submitted by the student and the committee shall decide whether to permit the student(s) to avail the Gap Year or not

6. Transitory Regulations

Discontinued, detained, or failed candidates are eligible for readmission as and when the semester is offered after fulfilment of academic regulations. Candidates who have been detained for want of attendance or not fulfilled academic requirements or who have failed after having undergone the course in earlier regulations or have discontinued and wish to continue the course are eligible for admission into the unfinished semester from the date of commencement of class work with the same or equivalent subjects as and when subjects are offered, subject to Section 2 and they will follow the academic regulations into which they are readmitted.

Candidates who are permitted to avail Gap Year shall be eligible for re-joining into the succeeding year of their B.CA from the date of commencement of class work, subject to Section 2 and they will follow the academic regulations into which they are readmitted.

7. Minimum Instruction Days for a Semester:

The minimum instruction days including exams for each semester shall be 90 days.

8. Medium of Instruction:

The medium of instruction of the entire B.C.A. / B.C.A. (Hons) programme will be in English only.

9. Student Transfers:

Student transfers shall be as per the guidelines issued by the Government of Andhra Pradesh and the Universities from time to time.

10. General Instructions:

i. The academic regulations should be read as a whole for purpose of any interpretation.
ii. Malpractices rules-nature and punishments are appended.
iii. Where the words “he”, “him”, “his”, occur in the regulations, they also include “she”, “her”, “hers”, respectively. iv. In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor is final.
v. The Institution may change or amend the academic regulations or syllabi at any time and the changes or amendments shall be made applicable to all the students on rolls with effect from the dates notified by the Institution.

vi. In the case of any doubt or ambiguity in the interpretation of the guidelines given, the decision of the Principal / Academics Council is final.

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MALPRACTICES RULES

DISCIPLINARY ACTION FOR / IMPROPER CONDUCT IN EXAMINATIONS

	Nature of Malpractices/Improper conduct	**Punishment**
	If the candidate:	
1. (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the University.
3.	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred and forfeits the seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the

		imposter is an outsider, he will be handed over to the police and a case is registered against him.
4.	Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the Chief Superintendent/Assistant – Superintendent / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in charge or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The candidates also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already

	outside the examination hall.	appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
8.	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat.
9.	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat. Person(s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the candidate has appeared including practical examinations and project work of that semester/year examinations.
12.	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the University for further action to award suitable punishment.	

Malpractices identified by squad or special invigilators

1. Punishments to the candidates as per the above guidelines.
2. Punishment for institutions: (if the squad reports that the college is also involved in encouraging malpractices)
   (i) A show cause notice shall be issued to the college.
   (ii) Impose a suitable fine on the college.
   (iii) Shifting the examination centre from the college to another college for a specific period of not less than one year.

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Structure of the Undergraduate Programme

All courses offered for the undergraduate program (B.C.A / B.C.A (Hons.)) are broadly classified as follows:

S.No.	Category	Breakup of Credits (Total 120)	Breakup of Credits (Total 160)	Statutory Body Recommendation	
				Breakup of Credits (Total 120)	Breakup of Credits (Total 160)
1	Major (Core)			60	80
2	Minor (Stream)			24	32
3	Multidisciplinary (MD)	12.5	12.5	9	9
4	Ability Enhancement Courses (AEC)	8.5	8.5	9	9
5	Skill Enhancement Courses (SEC)			8	8
6	Value Added Courses (VAC)			6-8	6-8
7	Summer Internship	2	2	2-4	2-4
8	Research / Dissertation	2	12	-	12

COURSE STRUCTURE

I SEMESTER										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	AEC-1	24BC1T01	Communicative English	3	0	0	3	30	70	100
2	VAC	24BC1T02	Computer Fundamentals	3	0	0	3	30	70	100
3	MD	24BC1T03	Fundamentals of Commerce	2	0	0	2	30	70	100
4	CORE	24BC1T04	Digital Logic Design	3	0	0	3	30	70	100
5	CORE	24BC1T05	Programming in C	3	0	0	3	30	70	100
6	MD	24BC1T06	Numerical and Statistical Methods	3	0	0	3	30	70	100
7	AEC-2	24BC1L01	Communication Skills Lab	0	0	3	1.5	30	70	100
8	VAC	24BC1L02	Office Automation Tools Lab	0	0	3	1.5	30	70	100
9	CORE	24BC1L03	Programming in C Lab	0	0	3	1.5	30	70	100
10	MD	24BC1L04	Statistics using R Lab	0	0	3	1.5	30	70	100
Total				**17**	**0**	**12**	**23**	**300**	**700**	**1000**

II SEMESTER										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	MD	24BC2T01	Discrete Mathematics	3	0	0	3	30	70	100
2	VAC	24BC2T02	Environmental Science	2	0	0	2	30	70	100
3	MD	24BC2T03	Business Organization	3	0	0	3	30	70	100
4	CORE	24BC2T04	Computer Organization	3	0	0	3	30	70	100
5	CORE	24BC2T05	Data Structures	3	0	0	3	30	70	100
6	CORE	24BC2T06	OOPs through C++	3	0	0	3	30	70	100
7	CORE	24BC2L01	Data Structures through C Lab	0	0	3	1.5	30	70	100
8	CORE	24BC2L02	OOPs through C++ Lab	0	0	3	1.5	30	70	100
9	VAC	24BC2T07	Indian Constitution	2	0	0	2	30	70	100
			Total	**19**	**0**	**6**	**22**	**270**	**630**	**900**

III SEMESTER										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	CORE	24BC3T01	Operating Systems	3	0	0	3	30	70	100
2	CORE	24BC3T02	Python Programming	3	0	0	3	30	70	100
3	CORE	24BC3T03	Database Management Systems	3	0	0	3	30	70	100
4	CORE	24BC3T04	Artificial Intelligence	3	0	0	3	30	70	100
5	CORE	24BC3L01	Operating Systems Lab	0	0	3	1.5	30	70	100
6	CORE	24BC3L02	Python Programming Lab	0	0	3	1.5	30	70	100
7	CORE	24BC3L03	Database Management Systems Lab	0	0	3	1.5	30	70	100
8	SEC	24BC3L04	Animation Design	0	0	3	1.5	30	70	100
9	SEC	24BC3Z01	Design Thinking and Innovation	0	1	2	2	30	70	100
Total				**13**	**1**	**12**	**20**	**270**	**630**	**900**

IV SEMESTER										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	CORE	24BC4T01	Computer Networks	3	0	0	3	30	70	100
2	SEC	24BC4T02	JAVA Programming	3	0	0	3	30	70	100
3	CORE	24BC4T03	Web Technologies	3	0	0	3	30	70	100
4	CORE	24BC4T04	Software Engineering	3	0	0	3	30	70	100
5	CORE	24BC4L01	Computer Networks Lab	0	0	3	1.5	30	70	100
6	SEC	24BC4L02	Android APP Development	0	0	3	1.5	30	70	100
7	SEC	24BC4L03	JAVA Programming Lab	0	0	3	1.5	30	70	100
8	SEC	24BC4L04	Web Technologies Lab	0	0	3	1.5	30	70	100
9	SI	24BC4I01	Summer Internship and Seminar*	-	-	-	2	50	-	50
Total				**12**	**0**	**12**	**20**	**290**	**560**	**850**

* Evaluated during the V Sem

V SEMESTER (Tentative)										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	CORE	24BC5T01	Professional Core -1	3	0	0	3	30	70	100
2	CORE	24BC5T02	Professional Core -2	3	0	0	3	30	70	100
3	Minor	24BC5E05	Minor Theory - 5	3	0	0	3	30	70	100
4	Minor	24BC5E06	Minor Theory - 6	3	0	0	3	30	70	100
5	CORE	24BC5L01	Professional Core -1 Lab	0	0	3	1.5	30	70	100
6	CORE	24BC5L02	Professional Core -2 Lab	0	0	3	1.5	30	70	100
7	SEC	24BC5S01	Skill Enhancement Lab - 3	0	1	2	2	30	70	100
8	Core	24BC5C01	Community Service Project	0	0	0	2	0	0	100
Total				**12**	**1**	**8**	**19**	**210**	**490**	**800**

VI SEMESTER (Tentative)										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	CORE	24BC6T01	Professional Core -1	3	0	0	3	30	70	100
2	CORE	24BC6T02	Professional Core -2	3	0	0	3	30	70	100
4	Minor	24BC6E07	Minor Theory - 7	3	0	0	3	30	70	100
5	Minor	24BC6E08	Minor Theory - 8	3	0	0	3	30	70	100
6	Core	24BC6P01	Major Project [Initiated in 5th Semester]	0	0	8	4	60	140	200
Total				**12**	**0**	**8**	**16**	**180**	**420**	**600**

VII SEMESTER (Tentative)										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	CORE	24BC7T01	Professional Core -1	3	0	0	3	30	70	100
2	CORE	24BC7T02	Professional Core -2	3	0	0	3	30	70	100
3	CORE	24BC7T03	Professional Core -3	3	0	0	3	30	70	100
4	Minor	24BC7E09	Minor Theory - 9	3	1	0	4	30	70	100
5	CORE	24BC7L01	Professional Core -1 Lab	0	0	3	1.5	30	70	100
6	CORE	24BC7L02	Professional Core -2 Lab	0	0	3	1.5	30	70	100
7	AEC-3	24BC7L03	Ability Enhancement Courses (AEC) -3	0	1	2	2	30	70	100
8	AEC-4	24BC7L04	Ability Enhancement Courses (AEC) -4	0	0	2	1	30	70	100
Total				**12**	**2**	**10**	**19**	**240**	**560**	**800**

VIII SEMESTER (Tentative)										
S.No	**Category**	**Code**	**Courses**	**L**	**T**	**P**	**C**	**IM**	**EM**	**TM**
1	CORE	24BC8T01	Professional Core -1	3	0	0	3	30	70	100
2	CORE	24BC8T02	Professional Core -2	3	0	0	3	30	70	100
3	Minor	24BC8E10	Minor Theory - 10	3	1	0	4	30	70	100
4	CORE	24BC8L01	Professional Core -1 Lab	0	0	3	1.5	30	70	100
5	CORE	24BC8L02	Professional Core -2 Lab	0	0	3	1.5	30	70	100
6	CORE	24BC8P02	Dissertation work	0	0	16	8	60	140	200
Total				**9**	**1**	**22**	**21**	**210**	**490**	**700**

I Semester	L	T	P	C
	3	0	0	3
24BC1T01 - COMMUNICATIVE ENGLISH				

Course Objectives:
The main objective of introducing this course, Communicative English, is to facilitate effective listening, Reading, Speaking and Writing skills among the students. It enhances the same in their comprehending abilities, oral presentations, reporting useful information and providing knowledge of grammatical structures and vocabulary. This course helps the students to make them effective in speaking and writing skills and to make them industry ready.

Course Outcomes(Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Identify and recall specific pieces of information through listening and reading texts.	K1
CO2	Demonstrate the ability to communicate ideas and concepts effectively through speaking and writing.	K2
CO3	Apply knowledge of grammatical structures and vocabulary in forming coherent sentences and paragraphs.	K3
CO4	Analyze and interpret texts by identifying key elements, context clues, and supporting ideas.	K4
CO5	Create structured essays, summaries, and formal presentations on specific topics using proper grammar.	K6

UNIT I

Lesson: HUMAN VALUES: Gift of Magi (Short Story)

Listening: Identifying the topic, the context and specific pieces of information by listening to short audio texts and answering a series of questions.

Speaking: Asking and answering general questions on familiar topics such as home, family, work, studies and interests; introducing oneself and others.

Reading: Skimming to get the main idea of a text; scanning to look for specific pieces of information.

Writing: Mechanics of Writing-Capitalization, Spellings, Punctuation-Parts of Sentences.

Grammar: Parts of Speech, Basic Sentence Structures-forming questions

Vocabulary: Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words

UNIT II

Lesson: NATURE: The Brook by Alfred Tennyson (Poem)

Listening: Answering a series of questions about main ideas and supporting ideas after listening to audio texts.

Speaking: Discussion in pairs/small groups on specific topics followed by short structure talks.

Reading: Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together.

Writing: Structure of a paragraph - Paragraph writing (specific topics)

Grammar: Cohesive devices - linkers, use of articles and zero article; prepositions.

Vocabulary: Homonyms, Homophones, Homographs.

UNIT III

Lesson: BIOGRAPHY: Elon Musk

Listening: Listening for global comprehension and summarizing what is listened to.

Speaking: Discussing specific topics in pairs or small groups and reporting what is discussed.

Reading: Reading a text in detail by making basic inferences -recognizing and interpreting specific context clues; strategies to use text clues for comprehension.

Writing: Summarizing, Note-making, paraphrasing

Grammar: Verbs - tenses; subject-verb agreement; Compound words, Collocations

Vocabulary: Compound words, Collocations

UNIT IV

Lesson: INSPIRATION: The Toys of Peace by Saki

Listening: Making predictions while listening to conversations/transactional dialogues without video; listening with video.

Speaking: Role plays for practice of conversational English in academic contexts(formal and informal) - asking for and giving information/directions.

Reading: Studying the use of graphic elements in texts to convey information, reveal trends/patterns/relationships, communicate processes or display complicated data.

Writing: Letter Writing: Official Letters, Resumes

Grammar: Reporting verbs, Direct & Indirect speech, Active & Passive Voice

Vocabulary: Words often confused, Jargons

UNIT V

Lesson: MOTIVATION: The Power of Intrapersonal Communication (An Essay)

Listening: Identifying key terms, understanding concepts and answering a series of relevant questions that test comprehension.

Speaking: Formal oral presentations on topics from academic contexts

Reading: Reading comprehension.

Writing: Writing structured essays on specific topics.

Grammar: Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement)

Vocabulary: Technical Jargons

Textbooks:

1. Pathfinder: Communicative English for Undergraduate Students, 1st Edition, Orient Black Swan,2023 (Units 1,2 & 3)
2. Empowering with Language by Cengage Publications, 2023 (Units 4 & 5)

Reference Books:

1. Dubey, Sham Ji& Co. English for Engineers, Vikas Publishers, 2020
2. Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge, 2014.
3. Murphy, Raymond. English Press, 2019.Grammar in Use, Fourth Edition, Cambridge University
4. Lewis, Norman. Word Power Made Easy- The Complete Handbook for Building a Superior

Vocabulary. Anchor, 2014.

Web Resources:

GRAMMAR:

1. www.bbc.co.uk/learningenglish
2. https://dictionary.cambridge.org/grammar/british-grammar/
3. www.eslpod.com/index.html
4. https://www.learngrammar.net/
5. https://english4today.com/english-grammar-online-with-quizzes/
6. https://www.talkenglish.com/grammar/grammar.aspx

VOCABULARY

1. https://www.youtube.com/c/DailyVideoVocabulary/videos
2. https://www.youtube.com/channel/UC4cmBAit8i_NJZE8qK8sfpA

I Semester	L	T	P	C
	3	0	0	3
24BC1T02 - COMPUTER FUNDAMENTALS				

Course Objectives:
The main objective of this course is to introduce the fundamentals of computing devices and the Internet. It focuses on computer literacy that prepares students for life-long learning of computer concepts and skills.

Course Outcomes(Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Identify the components of a computer system, types of software, and the basics of hardware and software.	K1
CO2	Explain the basic operations of a GUI-based operating system and commonly used software applications.	K2
CO3	Demonstrate the ability to operate MS Office tools (Word, Excel, PowerPoint) for various tasks.	K3
CO4	Analyze the concepts of the Internet, web browsers, computer networks, and their functions.	K4
CO5	Evaluate and apply knowledge of Google tools, Internet security, and electronic payment systems.	K5

Unit I:
Knowing computer: computer definition, applications of computer, components of a computer system, Central Processing Unit(CPU), Video Display Unit, Keyboard and Mouse, Optical Storage devices, basics of hard drive, concepts of hardware and software, concept of computing, data and information, applications of information electronics and communication technology, connecting keyboard, mouse, monitor and printer to CPU and checking power supply. **Computer software & its types**: System software, application software, types of operating system, role of operating system, utility programs, packages, communication software, commonly used application software

Unit II:

Operating computer using GUI based operating system: operating systems basics, basics of popular operating systems, the user interface, basics of operating system setup, common utilities
MS Windows Operating System: definition and functions, basic components of Windows, Icons, Desktop, Taskbar, Notification area, files and folders, start menu operations, my computer, network neighborhood, recycle bin, windows explorer, setting wall paper, creating, copying, moving and deleting files, changing the mouse pointer, paint, notepad, setting date and time, screen saver, using mouse: using right mouse button moving icons on the screen, use of common icons, status bar, using menu and menu selections, running an application, viewing of files, folder sand directories, creating and renaming of files and folders, opening and closing of different windows, using help, crating shortcuts, using Windows accessories.

Unit III:
MS-Word: introduction, Windows interface, document views, creating & editing documents, selecting, deleting, replacing text, copying text to another file, formatting text, paragraphs, fonts, paragraph formatting using bullets and numbering, dialog box, checking spellings, line spacing, margins, space before and after paragraph, mail merge, macros, printing documents, print preview.
MS Excel: introduction, workbook, worksheet, formatting, formulas, charts

MS Power point: Introduction, creating presentation with all options, slide show, animations and designs

Unit IV:

Introduction to Internet, WWW, Web browsers Basics of computer networks, LAN, WAN, concept of Internet, web browsing software, search engines, understanding URLs, domain name, IP address, Basics of electronic mail, getting and email account, sending and receiving mails, documentcollaboration

Unit V:

Useful Google tools such as drive, sheet, doc, meet etc., Firewall, computer virus, Anti-virus software, Internet security and privacy, Basics of Electronic Data Interchange, Electronic Payment systems, Types of payment systems, digital cash, electronic cheque, smart card, introduction to digital signatures.

Text Books:

1. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.
2. Computer Fundamentals, A. Goel, 2010, Pearson Education.
3. Fundamentals of Computers, P. K.Sinha& P. Sinha, 2007, BPB Publishers.
4. Fundamentals of Information Technology: Alexis Leon & Mathews Leon, Vikas Publishing House, NewDelhi.
5. IT Tools, R.K. Jain, Khanna Publishing House.
6. Introduction to Information Technology, Satish Jain, AmbrishRai&Shashi Singh, Paperback Edition, BPB Publications, 2014

Web Resources:

1. https://edu.gcfglobal.org/en/computerbasics/
2. https://edu.gcfglobal.org/en/subjects/office/
3. https://onlinecourses.swayam2.ac.in/nou20_cs03/
4. https://tutorialpoint.com/computer_fundamentals/index.htm
5. https://nptel.ac.in/courses/106/103/106103068/

I Semester	L	T	P	C
	2	0	0	2
24BC1T03 - FUNDAMENTALS OF COMMERCE				

Course Objectives:

The objective of this paper is to help students to acquire conceptual knowledge of the Commerce, Economy and Role of Commerce in Economic Development. To acquire Knowledge on Accounting and Taxation.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Define the basic concepts of commerce and explain its role in economic and societal development.	K1
CO2	Understand and apply key economic theories, including macro and microeconomics, demand, supply, and market structures.	K2
CO3	Demonstrate knowledge of accounting principles, branches of accounting, and the accounting cycle.	K3
CO4	Analyze different types of taxes, tax systems, and the role of taxation authorities like CBDT and CBIC.	K4
CO5	Evaluate the use of digital marketing, SEO, and data analytics in modern commerce and web development.	K5

Unit I:
Introduction: Definition of Commerce, Role of Commerce in Economic Development, Role Commerce in Societal Development. Imports and Exports, Balance of Payments. World Trade Organization.

Unit II:
Economic Theory: Macro Economics, Meaning, Definition, Measurements of National Income, Concepts of National Income. Micro Economics, Demand and Supply. Elasticity of Demand and Supply. Classification of Markets, Perfect Competition, Characteristics, Equilibrium Price, Marginal Utility.

Unit III:
Accounting Principles: Meaning and Objectives Accounting, Accounting Cycle, Branches of Accounting, Financial Accounting, Cost Accounting, Management Accounting. Concepts and Conventions of Accounting – GAAP.

Unit IV:
Taxation: Meaning of Tax, Taxation, Types of Tax, Income Tax, Corporate Taxation, GST, Customs & Exercise. Differences between Direct and Indirect Tax, Objectives of Tax, Concerned authorities, Central Board of Direct Taxes (CBDT) and Central Board of Excise and Customs (CBIC).

Unit V:
Computer Essentials: Web Design, Word Press Basics, Developing a Simple Website. Digital Marketing, Social Media Marketing, Content Marketing, Search Engine Optimization(SEO), E-mail Marketing. Data Analytics, Prediction of customer behavior, customized suggestions.

Reference Books:

1. S.P. Jain & K.L Narang, Accountancy - I Kalyani Publishers.
2. R.L. Gupta & V.K. Gupta, Principles and Practice of Accounting, Sultan Chand
3. Business Economics -S.Sankaran, Margham Publications, Chennai.
4. Business Economics - Kalyani Publications.
5. Dr. Vinod K. Singhania: Direct Taxes – Law and Practice, Taxmann Publications.
6. Dr. Mehrotra and Dr. Goyal: Direct Taxes – Law and Practice, Sahitya Bhavan Publications

I Semester	L	T	P	C
	3	0	0	3
24BC1T04 - DIGITAL LOGIC DESIGN				

Course Objectives:
The main objectives of the course are to

- Understand different methods used for the simplification of Boolean functions and binary arithmetic.
- Design and implement combinational circuits, synchronous & asynchronous sequential circuits.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Identify and recall the fundamentals of binary systems, number conversions, binary codes, and logic gates.	K1
CO2	Apply Boolean algebra theorems and simplification techniques to minimize Boolean functions using K-Maps and tabulation methods.	K3
CO3	Analyze and design combinational circuits like adders, subtractors, multiplexers, and comparators.	K4
CO4	Demonstrate the working of synchronous sequential circuits, including flip-flops, state reduction, and clocked circuits.	K3
CO5	Design and evaluate asynchronous sequential circuits and counters, including registers and latches.	K5

UNIT I:
Binary Systems: Digital Systems, Binary Numbers, Number Base Conversions, Octal and Hexadecimal Numbers, Complements, Signed Binary Numbers, Binary Codes, Binary Storage and Registers, Binary Logic.
Boolean Algebra and Logic Gates: Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, Digital Logic Gates.

UNIT II:
Minimization: K-Map Method – Tabulation Method, POS - SOP, Don't Care Conditions, NAND, NOR Implementation.
Combinational Logic: Combinational Circuits, Analysis and Design Procedure, Binary Adder, Subtractor, Decimal Adder, Binary Multiplier, Magnitude Comparator, Decoders, Encoders, Multiplexers.

UNIT III:
Synchronous Sequential Logic: Sequential Circuits - Latches, Flip-Flops, An analysis of Clocked Sequential Circuits, State Reduction and Assignment Design Procedure.

UNIT IV:
Registers and Counters: Registers, Shift Registers, Ripple Counters, Synchronous Counters, Ring Counters-Johnson Counter.

UNIT V:
Asynchronous Sequential Circuit: Introduction, Analysis Procedure, Circuits with Latches, Design Procedure.

Text Books:

1. M. Morris Mano, "Digital Design", 3rd edition, Pearson Education, Delhi, 2007.
2. Carl Hamacher, Z. Vranesic, S. Zaky: Computer Organization, 5/e (TMH).

Reference Books:

1. Donald P Leech, Albert Paul Malvino and GoutamSaha,"Digital Principles and Applications", Tata McGraw Hill, 2007.

Web Resources:

1. https://nptel.ac.in/courses/117105080
2. Digital Logic Design: Theory (CS226) + Lab (CS254) (iitb.ac.in)
3. Free Course: Digital Logic and Circuits Simulations from Banaras Hindu University | Class Central

I Semester	L	T	P	C
	3	0	0	3
24BC1T05 - PROGRAMMING IN C				

Course Objectives:

The main objectives of the course is to

- Introduce students to the fundamentals of computer programming.
- Provide hands-on experience with coding and debugging.
- Foster logical thinking and problem-solving skills using programming.
- Familiarize students with programming concepts such as data types, control structures, functions, and arrays. Encourage collaborative learning and teamwork in coding projects.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the fundamentals of computer programming, including algorithms, flowcharts, and basic data types.	K2
CO2	Apply control structures like conditional statements, loops, and unconditional statements to solve programming problems.	K3
CO3	Implement programs using arrays and strings for efficient data handling and manipulation.	K3
CO4	Analyze the use of pointers and user-defined data types like structures and unions in complex programs.	K4
CO5	Develop programs using functions, recursion, and file handling techniques to solve real-world problems.	K5

UNIT I:

Introduction to Programming and Problem Solving: History of Computers, Basic organization of a computer: ALU, input-output units, memory, program counter, Introduction to Programming Languages, Basics of a Computer Program- Algorithms, flowcharts (Using Dia Tool), pseudo code. Introduction to Compilation and Execution, Primitive Data Types, Variables, and Constants, Basic Input and Output, Operations, Type Conversion, and Casting. Problem solving techniques: Algorithmic approach, characteristics of algorithm, Problem solving strategies: Top-down approach, Bottom-up approach, Time and space complexities of algorithms.

UNIT II :

Control Structures: Simple sequential programs Conditional Statements (if, if-else, switch), Loops (for, while, do-while) and Unconditional statements: goto, Break and Continue.

UNIT III:

Arrays and Strings: Arrays indexing, memory model, programs with array of integers, two dimensional arrays, Introduction to Strings and string Operations

UNIT IV:

Pointers & User Defined Data types: Pointers, dereferencing and address operators, pointer and address arithmetic, array manipulation using pointers, User-defined data types-Structures and

Unions.

UNIT V:

Functions & File Handling: Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, modifying parameters inside functions using pointers, arrays as parameters, Recursive functions. Scope and Lifetime of Variables, Basics of File Handling

Text Books:

1. E. Balagurusamy,"Programming in ANSI C", 4/e, TMH
2. Dr N B Venlateswarlu," C Programming", S Chand Publications
3. YashwantKanetkar,"Let Us C: Authentic guide to C programming language", 19th Edition, BPB publications
4. B. Kernighan & Dennis Ritchie, "The C Programming Language", 2/e PHI

Reference Books:

1. Paul Deitel, Harvey Deitel, "C: How to Program", 8/e, Prentice Hall.
2. Schaum's Outline of Education, 1996 Programming with C,Byron S Gottfried,McGraw-Hill

Web Resources:

1. Introduction to Programming in C - Course (nptel.ac.in)
2. Lecture Notes | Practical Programming in C | Electrical Engineering and Computer Science | MIT OpenCourseWare
3. https://nptel.ac.in/courses/106105171
4. http://cslibrary.stanford.edu/

I Semester	L	T	P	C
	3	0	0	3
24BC1T06 - NUMERICAL AND STATISTICAL METHODS				

Course Objectives:
The main objectives of the course are:

- To learn how to perform error analysis for arithmetic operations.
- To demonstrate working of various numerical methods.
- To provide a basic understanding of the derivation and use of methods of interpolation and numerical integration.
- To impart knowledge of various statistical techniques.
- To develop students understanding through laboratory activities to solve problems related to above stated concepts.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Apply methods like Secant, Regula-Falsi, Newton-Raphson, and Fixed Point Iteration to solve equations and perform error analysis.	K3
CO2	Solve systems of linear equations using methods like Gauss Elimination, Gauss-Jordan, Gauss-Seidel, and LU-Decomposition, and compute Eigenvalues and Eigenvectors.	K3
CO3	Apply interpolation techniques (Newton's and Lagrange's methods) and numerical methods like Trapezoidal and Simpson's for differentiation and integration.	K3
CO4	Analyze data using statistical measures like mean, median, mode, standard deviation, skewness, kurtosis, and correlation coefficients.	K4
CO5	Understand and apply theorems of probability, including Bayes' Theorem and conditional probability, to solve problems.	K3

UNIT I:
Solution of equations (polynomial and transcendental equations) interval having methods, secant, Regula – Falsi, Newton – Raphson methods, Fixed point Iteration method.

UNIT II:
Solution of system of linear equations: Gauss – Elemination method,Gauss – Jordan,Gauss – Siedel iteration method,LU- Decomposition method,Eigen values and Eigen vectors of a square matrix.

UNIT III:
Interpolation: Forward and backward differences, Newton's forward and backward formula, Lagrange's interpolation and Lagrange's inverse interpolation formula. Numerical differentiation, integration: Numerical differentiation forward and backward formula, Trapezoidal and Simpsons formulas.

UNIT IV:
Basic concepts and definition of statistics : Mean, Median, Mode, standard deviation, coefficient of variation, skewness and kurtosis ,Karl Pearson Correlation coefficient , Rank Correlation and illustrated examples.

UNIT V:

Probability: Basic concepts and definition of probability, Probability axioms, Conditional probability, Addition and Multiplication theorem of probability (Based on set theory concepts), Bayes theorem, problems and applications.

Text Books:

1. Sunil S .Patil Numerical and Statistical Methods EBPB.
2. S.S.Shastry Introductory methods of Numerical Analysis PHI (New Delhi).

Reference Books:

1. Gupta S.C & Kapuram VK Fundamentals of Mathematical Statistics.

I Semester	L	T	P	C
	0	0	3	1.5
24BC1L01 - COMMUNICATION SKILLS LAB				

Course Objectives:

The main objective of introducing this course, Communicative English Laboratory, is to expose the students to a variety of self-instructional, learner friendly modes of language learning. The students will get trained in basic communication skills and also make them ready to face job interviews.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the basics of vowels, consonants, and neutralization/accent rules to improve pronunciation and articulation.	K2
CO2	Apply effective communication skills, including JAM (Just A Minute) sessions and conversational practice for fluency in English.	K3
CO3	Develop professional writing skills through e-mail writing, resume creation, cover letter drafting, and writing a Statement of Purpose (SOP)	K4
CO4	Engage in group discussions and debates by applying various methods, enhancing critical thinking and argumentation skills.	K4
CO5	Demonstrate interview and presentation skills through role-play, poster/PPT presentations, and practicing interview techniques.	K3

List of Topics:

1. Vowels & Consonants
2. Neutralization/Accent Rules
3. Communication Skills & JAM
4. Role Play or Conversational Practice
5. E-mail Writing
6. Resume Writing, Cover letter, SOP
7. Group Discussions-methods & practice
8. Debates - Methods & Practice
9. PPT Presentations/ Poster Presentation
10. Interviews Skills

Suggested Software:

- Walden Infotech
- Young India Films

Reference Books:

1. Raman Meenakshi, Sangeeta-Sharma. Technical Communication. Oxford Press. 2018.
2. Taylor Grant: English Conversation Practice, Tata McGraw-Hill Education India, 2016
3. Hewing's, Martin. Cambridge Academic English (B2). CUP, 2012.
4. J. Sethi& P.V. Dhamija. A Course in Phonetics and Spoken English, (2nd Ed) , Kindle, 2013

Web Resources:

Spoken English:

1. www.esl-lab.com
2. www.englishmedialab.com
3. www.englishinteractive.net
4. https://www.britishcouncil.in/english/online
5. http://www.letstalkpodcast.com/
6. https://www.youtube.com/c/mmmEnglish_Emma/featured
7. https://www.youtube.com/c/ArnelsEverydayEnglish/featured
8. https://www.youtube.com/c/engvidAdam/featuredhttps://www.youtube.com/c/EnglishClass101/featured
9. https://www.youtube.com/c/EnglishClass101/featured
10. https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists
11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw

Voice & Accent:

1. https://www.youtube.com/user/letstalkaccent/videos
2. https://www.youtube.com/c/EngLanguageClub/featured
3. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc
4. https://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA

I Semester	L	T	P	C
	0	0	3	1.5
24BC1L02 - OFFICE AUTOMATION TOOLS LAB				

Course Objectives:

- To introduce the environment of GUI in Ms-Word and its features.
- To introduce the fundamental concepts using Ms-Word and its features to make it more useful.
- To provide hands-on use of Word, Excel and PowerPoint.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the basic components and peripherals of a computer, including the CPU, and perform tasks such as assembling/disassembling a PC and installing operating systems.	K2
CO2	Apply knowledge of basic office tools such as Microsoft Word, Excel, and PowerPoint to create and format professional documents, spreadsheets, and presentations.	K3
CO3	Develop skills in advanced document handling techniques such as mail merge, document formatting, and the use of tables for various professional purposes.	K4
CO4	Analyze and manage data in spreadsheets using formulas, sorting, filtering, and chart generation for effective data presentation and decision-making.	K4
CO5	Demonstrate effective presentation skills by creating presentations with transitions, animations, custom layouts, and the use of graphical elements to enhance content delivery.	K3

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Every student should disassemble and assemble the PC back to working condition.
Experience the following:
Binary Arithmetic (https://scratch.mit.edu/projects/613350058)
Full adder circuit (https://logic.ly/demo/ andhttps://logic.ly/demo/samples)
This app is to test unsigned binary numbers to decimal number conversion skill of the students.
https://scratch.mit.edu/projects/621571989

Task 3: Installation of MS windows on the personal computer.

Task 4: Search Engines & Netiquette: Students should know what search engines are and how to use the search engines. A few topics would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task 5: Creation of Gmail account, accessing Google Drive including Google Docs

Task 6: Create a document to write a letter to the DM&HO of the district complaining about Hygienic conditions in your area. (*Similar Letters)

Task 7: Create a document to share your experience of your recent vacation with family.

Task 8: Create a document to send a holiday intimation to all the parents at time about Dasara Vacation. (using mail merge)

Task 9: Create a document to create Time Table of you class using tables.

Task 10: Create a table with following columns and display the result in separate cells for the following

- Emp Name, Basic pay, DA, HRA, Total salary.
- Sort all the employees in ascending order with the name as the key
- Calculate the total salary of the employee
- Calculate the Grand total salary of the employee

Task 11: Create two pages of curriculum vitae of a graduate with the following specifications

- Table to show qualifications with proper headings
- Appropriate left and right margins
- Format ½pageusingtwo-columnapproachabout yourself
- Name on each page at the top right side
- Page no. in the footer on the right side

Task 12: Create a letter as the main document and create 10 records for the 10 persons. Use mail merge to create letter for selected persons among 10.

Task 13: Create a worksheet with you class marks displaying total, average, top marks in the class and least marks in the class.

Task 14: Create a Worksheet with employee no, name, job, salaries of 10 employees, calculate DA,TA,HRA ,Gross Salary and Net Salary.

I. Find the sum of HRA's of Total employees.
II. Find the average DA
III. Display the Maximum salary of the employee.

Task 15: Prepare a chart with height and weights of you class mates in at least 3 types of charts.

Task 16: Demonstrate the use of Filter with the attendance data of your class.

Task 17: Prepare a presentation with your achievements and experiences in College.

Task 18: Create a Presentation of your organization with pictures, clip arts and animations

Task 19: Create a presentation using templates.

Task 20: Create a Custom layout or Slide Master for professional presentation.

Task 21: Create a presentation with slide transitions and animation effects.

Task 22: Create a table in PPT and apply graphical representation

Text Books:

1. Computer Fundamentals–Pradeep .K. Sinha: BPB Publications.
2. Joy of Computing: ABC of bits, trits, qubits, bytes, trytes, qutrits, quids for kids to Teachers, N B V Venkateswarlu, 2022
3. Fundamentals of Computers –ReemaThareja, Oxford University Press India

Web Resources:

1. https://onlinecourses.swayam2.ac.in/cec21_cs15/preview
2. https://edu.gcfglobal.org/en/computerbasics/
3. https://www.computerhope.com/jargon/m/microsoft-word.htm
4. https://www.coursera.org/courses?query=office%20automation

I Semester	L	T	P	C
	0	0	3	1.5
24BC1L03 - PROGRAMMING in C LAB				

Course Objectives:

The course aims to give students hands on experience and train them on the concepts of the C-programming language.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the fundamentals of programming, including writing and executing basic C programs using arithmetic and input/output statements.	K2
CO2	Apply conditional statements, loops, and functions to solve computational problems involving decision-making and iterative processes.	K3
CO3	Analyze and manipulate arrays, strings, pointers, and structures in C to perform various operations and memory management.	K4
CO4	Implement complex programming constructs such as recursive functions, file handling, and dynamic memory allocation in C	K3
CO5	Demonstrate problem-solving skills using C programming to handle file operations, pointer arithmetic, and structure/union concepts.	K4

Suggested Experiments/Activities

WEEK 1: Lab1: Familiarization with programming environment

i) Basic Linux environment and its editors like Vi, Vim &Emacs etc.
ii) Exposure to Turbo C, gcc
iii) Writing simple programs using using input and Output Statements
iv) Simple Arithmetic Operations

WEEK 2: Lab2: Converting algorithms/flow charts into C Source code.
Developing the algorithms/flowcharts for the following sample programs

i) Simple statistics Operations- Sum and average etc
ii) Conversion of Fahrenheit to Celsius and vice versa
iii) Simple and Compound Interest calculation

WEEK 3: Lab 3: Simple computational problems using arithmetic expressions.

i) Finding the square root of a given number
ii) Finding compound interest
iii) Area of a triangle using hero's formulae
iv) Distance travelled by an object

WEEK 4: Lab 4: Simple computational problems using the operator precedence and associativity

i) Evaluate the following expressions.
   a. A+B*C+(D*E) + F*G
   b. A/B*C-B+A*D/3
   c. A+++B---A

d. J= (i++) + (++i)

ii) Find the maximum of three numbers using conditional operator

WEEK 5: Lab 5: Problems involving if-then-else structures.

i) Find the max and min of four numbers using if-else.
ii) Generate electricity bill.
iii) Find the roots of the quadratic equation.
iv) Simulate a calculator using switch case.
v) Find the given year is a leap year or not etc

WEEK 6: Lab 6: Iterative problems e.g., series and sequences

i) Find the factorial of given number using any loop.
ii) Find the given number is a prime or not.
iii) Checking a number is palindrome or not

WEEK 7: Lab 7:1D Array manipulation, linear search

i) Find the min and max of a 1-D integer array.
ii) Perform linear search on1D array.
iii) The reverse of a 1D integer array

WEEK 8: Lab 8: Matrix problems, String operations

i) Addition of two matrices
ii) Multiplication two matri es
iii) Sort array elements
iv) Concatenate two strings without built-in functions
v) Reverse a string using built-in and without built-in string functions

WEEK 9: Lab 9: Pointers and structures, memory dereference.

i) Write a C program to find the sum of a 1D array using malloc()
ii) Write a C program to find the total, average of n students using structures
iii) Enter n students data using calloc() and display failed students list

WEEK 10: Lab10 :

i) Demonstrate the differences between structures and unions using a C program.
ii) Write a C program to copy one structure variable to another structure of the same type.

WEEK 11: Lab 11: Write a C program to swap two numbers (Use call by value and call by reference)

WEEK 12: Lab 12: Recursive functions

i) Generate Fibonacci series.
ii) Find the LCM of two numbers.
iii) Find the factorial of a number.

WEEK 13: Lab 13: File operations

i) Write and read text into a file.
ii) Write and read text into a binary file

iii) Copy the contents of one file to another file.
iv) Find no. of lines, words and characters in a file

Text Books:

1. Ajay Mittal, Programming in C: A practical approach, Pearson.
2. Byron Gottfried, Schaum's Outline of Programming with C, McGraw Hill
3. Dr N B Venlateswarlu, " C Programming", S Chand Publications

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice- Hall of India
2. C Programming, A Problem-Solving Approach, Forouzan,

Web Resources:

1. Department of Computer Science and Technology – Course pages 2021–22: Programming in C and C++ – Course materials (cam.ac.uk)
2. Practical Programming in C | Electrical Engineering and Computer Science | MIT OpenCourseWare
3. Lab 1: Introduction to C | Systems, Networks, and Concurrency (anu.edu.au)
4. https://ps-iiith.vlabs.ac.in/
5. https://www.learn-c.org/

I Semester	L	T	P	C
	0	0	3	1.5
24BC1L04 - STATISTICS USING R LAB				

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand and use basic R programming constructs such as variables, data types, arithmetic operations, and input/output.	K2
CO2	Apply functions, vectors, matrices, and data frames in R for data manipulation and basic operations	K3
CO3	Analyze and implement object-oriented programming in R using S3, S4, and reference classes.	K4
CO4	Utilize R for statistical analysis, including calculating descriptive statistics and visualizing data using various types of plots (box plots, scatter plots, bar plots).	K4
CO5	Develop R programs to work with probability distributions, perform correlation analysis, and use R for basic statistical modeling.	K4

Exercise 0:

a) Installing R and R Studio Basic functionality of R

Exercise 1:

a) Write an R program to print Hello World
b) Write a program to demonstrate the basic Arithmetic in R
c) Write a program to demonstrate the Variable assignment in R

Exercise 2:

a) Write a program to demonstrate the data types in R
b) Write a program to demonstrate the creating and naming a vector in R
c) Write a program to demonstrate the create a matrix and naming matrix in R

Exercise 3:

a) Write a program to demonstrate the Add column and Add a Row in Matrix in R
b) Write a program to demonstrate the selection of elements in Matrixes in R
c) Write a program to demonstrate the Performing Arithmetic of Matrices

Exercise 4:

a) Write a program to demonstrate the Factors in R
b) Implement the Factor in R
c) Write a program to illustrate Ordered Factors in R

Exercise 5:

a) Write an R program to take input from the user.
b) Write an R program to Check if a Number is Odd or Even
c) Write an R program to check if the given number is a Prime Number

Exercise 6:

a) Write an R program to Find the Factorial of a Number
b) Write an R program to Find the Factors of a Number
c) Write an R program to Find the Fibonacci sequence Using Recursive Function

Exercise 7:

a) Write an R program to make a Simple Calculator
b) Write an R program to Find L.C.M of two numbers
c) Write an R program to create a Vector and to access elements in a vector

Exercise 8:

a) Write an R program to create a Matrix and access rows and columns using functions colnames() and rownames() .
b) Write an R Program to create a Matrix using cbind() and rbind() functions

Exercise 9:

a) Write an R Program to create a Matrix from a Vector using the dim() function.
b) Write an R Program to create a List and modify its components.
c) Write an R Program to create a Data Frame.

Exercise 10:

a) Write an R Program to access a Data Frame like a List.
b) Write an R Program to access a Data Frame like a Matrix.
c) Write an R Program to create a Factor.

Exercise 11:

a) Write an R Program to Access and Modify Components of a Factor.
b) Write an R Program to create an S3 Class and S3 Objects.
c) Write an R Program to write an own generic function in S3 Class.

Exercise 12:

a) Write an R Program to create an S4 Class and S4 Objects.
b) Write an R Program to write an own generic function in S4 Class.
c) Write an R Program to create a reference class and modify its methods.

Exercise 13:

a) Write an R program to create a scatter plot for the data frame columns.
b) Write an R program to create a bar plot for the data frame columns.
c) Write an R program to create a box plot for the data frame columns.

Exercise 14:

a) Write an R program to add the legend to the plot.
b) Write an R program to change the width and height of the plot layout
c) Write an R program to calculate mean, mode, median and standard devotion
d) Implement R Script to create a Pie chart, Bar Chart, scatter plot and Histogram.

Exercise 15:

a) Implement R Script to perform Normal, Binomial distributions.
b) Implement R Script to perform correlation

References:

1. R Cookbook Paperback – 2011 by Teetor Paul O Reilly Publications
2. Beginning R: The Statistical Programming Language by Dr. Mark Gardener, Wiley Publications
3. R Programming For Dummies by JorisMeysAndrie de Vries, Wiley Publications
4. Hands-On Programming with R by Grolemund, O Reilly Publications
5. Statistical Programming in R by KG Srinivas G.M. Siddesh, ChetanShetty&Sowmya B.J. - 2017 edition

6. R Fundamentals and Programming Techniques, Thomas Lumely.
7. R for Everyone Advanced Series Analytics and Graphics, Jared P. Lander- Addison Wesley
8. The Art of R Programming, Norman Matloff, Cengage Learning

Web Links:

1. URL: https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf
2. http://nptel.ac.in/courses/106104135/48
3. http://nptel.ac.in/courses/110106064/

Software Requirements:

1. The R statistical software program. Available from: https://www.r-project.org/
2. RStudio an Integrated Development Environment (IDE) for R. Available from: https://www.rstudio.com/

II Semester	L	T	P	C
	3	0	0	3
24BC2T01 - DISCRETE MATHEMATICS				

Course Objectives:

- To learn the mathematical foundations for Computer Science.
- Topics covered essential for understanding various courses

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand and apply propositional and predicate calculus, including logical connectives, truth tables, and inference rules.	K2
CO2	Understand and apply basic set theory concepts, including relations, functions, and the principle of inclusion and exclusion.	K2
CO3	Apply combinatorial techniques (permutations, combinations) and solve recurrence relations using generating functions.	K3
CO4	Understand and apply fundamental graph theory concepts, including graph representations, connectivity, and traversal algorithms.	K3
CO5	Understand and apply finite state machines, including DFA, NFA, and the pumping lemma for regular languages.	K2

Unit-I:
Mathematical Logic: Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof, Predicate Calculus: Predicates, Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

Unit-II:
Sets and Functions: Sets, Relations, Functions, Closures of Equivalence Relations, Partial ordering well ordering, Lattice, Sum of products and product of sums principle of Inclusions and Exclusions

Unit III:
Combinatory: Permutations, Combinations, Pigeonhole principle
Recurrence Relation: Linear and Non-linear Recurrence Relations, Solving Recurrence Relation using Generating Functions.

Unit IV:
Graphs: Introduction to graphs, graphs terminologies, Representation of graphs, Isomorphism
Connectivity & Paths: Connectivity, Euler and Hamiltonian Paths, Introduction to tree, tree traversals, spanning tree and tree searches: Breadth first search, Depth first search, cut-set, cut-vertex.

Unit-V:
Modeling Computation: Finite State Machine, Deterministic Finite Automata (DFA), Non- Deterministic Finite Automata Lemma for Regular Language.(NFA), Grammars and Language, Application of Pumping

Text Books:

1. Discrete Mathematics and its Applications with Combinatory and Graph Theory, 7th edition by Kenneth H. Rosen.
2. Discrete Mathematical Structures with Applications to Computer Science, J. P Tremblay, R. Manohar, TMH, 1997.
3. Elements of Discrete Mathematics -A Computer Oriented Approach, C. L. Liu P. Mohapatra, 3rd Edition, Tata McGraw Hill.
4. Discrete Mathematics, An open Introduction, Oscar Levin, 3rd edition.

Reference Books:

1. Elements of Discrete Mathematics by C. L. Liu and D.P. Mohapatra, TMH, 2012.
2. A Modern Approach to Discrete Mathematics and Structure by J. K. Mantri& T. KTripathy,Laxmi Publication.

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc22_cs123/preview
2. https://discrete.openmathbooks.org/preview/
3. https://mathworld.wolfram.com/topics/DiscreteMathematics.html
4. https://www.csie.ntu.edu.tw/~sylee/courses/dm/resources.htm

II Semester	L	T	P	C
	2	0	0	2
24BC2T02 - ENVIRONMENTAL SCIENCE				

Course Objectives:

- To make the students to get awareness on environment
- To understand the importance of protecting natural resources, ecosystems for future generations and pollution causes due to the day to day activities of human life
- To save earth from the inventions by the engineers.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand and analyze the scope, importance, and associated problems of natural resources, including renewable and non-renewable resources, and their impact on the environment.	K2
CO2	Understand the structure and function of different ecosystems and the importance of biodiversity, including conservation strategies and threats to biodiversity.	K2
CO3	Identify and evaluate the causes, effects, and control measures of various types of environmental pollution and solid waste management.	K3
CO4	Analyze social issues related to environmental sustainability, including urban problems, climate change, and the enforcement of environmental legislation.	K4
CO5	Understand the impact of human population dynamics on the environment and health, and explore the role of technology in addressing these issues.	K3

UNIT–I

Multidisciplinary Nature Of Environmental Studies: Definition, Scope and Importance, Need for Public Awareness. Natural Resources: Renewable and non-renewable resources, Natural resources and associated problems, Forest resources, Use and over, exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people, Water resources, Use and over utilization of surface and ground water, Floods, drought, conflicts over water, dams–benefits and problems, Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies: Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies: Energy resources.

UNIT–II

Ecosystems: Concept of an ecosystem. Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem, Ecological succession, Food chains, food webs and ecological pyramids: Introduction, types,characteristic features, structure and function of the following ecosystem:

a. Forest ecosystem.
b. Grassl and ecosystem
c. Desert ecosystem
d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Biodiversity And Its Conservation: Introduction Definition: genetic, species and ecosystem diversity– Bio-

geographical classification of India, Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values, Biodiversity at global, National and local levels, India as a mega-diversity nation, Hot-sports of biodiversity, Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts, Endangered and endemic species of India, Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT–III

Environmental Pollution: Definition, Cause, effects and control measures of:

a. Air Pollution.
b. Water pollution
c. Soil pollution
d. Marine pollution
e. Noise pollution
f. Thermal pollution
g. Nuclear hazards

Solid Waste Management: Causes, effects and control measures of urban and industrial wastes, Role of an individual in prevention of pollution, Pollution case studies, Disaster management: floods, earthquake, cyclone and landslides.

UNIT–IV

Social Issues and the Environment: From Unsustainable to Sustainable development, Urban problems related to energy, Water conservation, rain water harvesting, watershed management, Resettlement and rehabilitation of people; its problems and concerns. Case studies: Environmental ethics: Issues and possible solutions, Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies: Wastel and reclamation. Consumerism and waste products. Environment Protection Act, Air (Prevention and Control of Pollution) Act.Water (Prevention and control of Pollution) Act, Wild life Protection Act, Forest Conservation Act, Issues involved in enforcement of environmental legislation, Public awareness.

UNIT–V

Human Population And The Environment: Population growth, variation among nations. Population explosion, Family Welfare Programmes. Environment and human health, Human Rights, Value Education, HIV/AIDS, Women and Child Welfare, Role of information Technology in Environment and human health–Case studies. Field Work: Visitto a local area to document environmental assets River/ forest grassland/ hill/ mountain, Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds– river, hills lopes, etc.

Text books:

1. Text book of Environmental Studies for Undergraduate Courses ErachBharucha for University Grants Commission ,Universities Press.
2. Palaniswamy, “Environmental Studies”, Pearson education
3. S. AzeemUnnisa,“ Environmental Studies” Academic Publishing Company
4. K. RaghavanNambiar,“ Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, Scitech Publications (India), Pvt. Ltd.

Reference Books:

1. Deeksha Dave and E. Sai Baba Reddy, "Text book of Environmental Science", Cengage Publications.
2. M. Anji Reddy, "Text book of Environmental Sciences and Technology", B S Publication.
3. J.P.Sharma, Comprehensive Environmental studies, Laxmi publications.
4. J.Glynn Henry and Gary W. Heinke, "Environmental Sciences and Engineering",Prentice Hall of India Private limited
5. G.R.Chatwal, "A Text Book of Environmental Studies "Himalaya Publishing House.
6. Gilbert M. Masters and Wendell P.Ela, "Introduction to Environmental Engineering and Science,Prentice

II Semester	L	T	P	C
	3	0	0	3
24BC2T03 - BUSINESS ORGANIZATION				

Course Objectives:

The course aims to acquire conceptual knowledge of business, formation various business organizations. To provide the knowledge on deciding plant location, plan layout and business combinations.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the fundamental concepts of business, including its features, stages of development, and classification of business activities, as well as differentiate between industry, commerce, business, and profession.	K2
CO2	Analyze the considerations for establishing a new business, understand the qualities of a successful businessman, and evaluate the different forms of business organization, including their merits and demerits.	K4
CO3	Evaluate the factors affecting plant location and layout, understand the objectives and types of plant layouts, and assess the criteria and factors influencing the size of a business unit and its optimal size.	K4
CO4	Understand the concepts of business combination and rationalization, including their characteristics, objectives, causes, forms, and differences between rationalization and nationalization.	K4
CO5	Understand the evolution of computers, internet basics, network security concepts, and ethical implications in information technology.	K4

Unit I:
Business: Concept, Meaning, Features, Stages of development of business and importance of business. Classification of Business Activities. Meaning, Characteristics, Importance and Objectives of Business Organization. Difference between Industry & Commerce and Business & Profession, Modern Business and their Characteristics.

Unit II:
Promotion of Business: Considerations in Establishing New Business. Qualities of a Successful Businessman. Forms of Business Organization - Sole Proprietorship, Partnership, Joint Stock Companies & Co-operatives and their Characteristics, relative merits and demerits, Difference between Private and Public, Concept of One Person Company.

Unit III:
Plant Location and Layout: Meaning, Importance, Factors affecting Plant Location. Plant Layout - Meaning, Objectives, Importance, Types of Layout. Factors affecting Layout. Size of Business Unit - Criteria for Measuring the Size and Factors affecting the Size. Optimum Size and factors determining the Optimum Size.

Unit IV:
Business Combination: Meaning, Characteristics, Objectives, Causes, Forms and Kinds of Business

Combination. Rationalization: Meaning, Characteristics, Objectives, Principles, Merits and demerits, Difference between Rationalization and Nationalization.

Unit V:

Computer Essentials: Milestones of Computer Evolution – Computer, Block diagram, generations of computer. Internet Basics - Internet, history, Internet Service Providers, Types of Networks, IP, Domain Name Services, applications. Ethical and Social Implications - Network and security concepts- Information Assurance Fundamentals, Cryptography-Symmetric and Asymmetric Malware, Firewalls, Fraud Techniques, privacy and data protection.

Reference Books:

1 Gupta, C.B., "Business Organisation", MayurPubliction, (2014).
2 Singh,B.P.,Chhabra,T.N.,"An Introduction to Business Organisation & Management", KitabMahal, (2014).
3 Sherlekar, S.A. &Sherlekar, V.S, "Modern Business Organization & Systems Approach Mumbai", Himalaya Publishing House, (2000).
4 Bhusan Y. K., "Business Organization", Sultan Chand & Sons.
5 Prakash, Jagdish, "Business Organistaton and Management", KitabMahal Publishers (Hindi and English)
6 Fundamentals of Computers by V. Raja Raman
7 Cyber Security Essentials by James Graham, Richard Howard, Ryan Olson

II Semester	L	T	P	C
	3	0	0	3
24BC2T04 - COMPUTER ORGANIZATION				

Course Objectives:

The purpose of the course is to introduce principles of computer organization and the basic architectural concepts. It provides an in depth understanding of basic organization, design, programming of a simple digital computer, computer arithmetic, instruction set design, micro programmed control unit, pipelining and vector processing, memory organization and I/O systems

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the basic structure of computers, including types, functional units, bus structures, software, performance factors, and data representation methods such as fixed-point and floating-point.	K2
CO2	Apply register transfer language to describe micro-operations including arithmetic, logic, and shift operations, and understand instruction codes, computer registers, and instruction cycles.	K3
CO3	Understand micro programmed control mechanisms and the organization of the central processing unit, including instruction formats, addressing modes, and data manipulation.	K4
CO4	Analyze different types of memory organization, including memory hierarchy, main memory, cache memory, and virtual memory, and understand concepts related to shift registers and RAID.	K4
CO5	Understand the organization of input-output systems, including peripheral devices, interfaces, data transfer modes, priority interrupts, DMA, and serial communication.	K2

UNIT-I:
Basic Structure Of Computers: Computer Types, Functional unit, Basic Operational concepts, Bus structures, Software, Performance, multiprocessors and multi computers. Data Representation. Fixed Point Representation. Floating – Point Representation.

UNIT-II:
Register Transfer Language And Micro-operations: Register Transfer language. Register Transfer Bus and memory transfers, Arithmetic Micro-operations, Logic micro operations, shift micro operations, Arithmetic logic shift unit. Instruction codes. Computer Registers, Computer instructions, Instruction cycle.

UNIT III:
Micro Programmed Control: Control memory, Address sequencing, micro program example, design of control unit.
Central Processing Unit: General Register Organization, Instruction Formats, Addressing modes, Data Transfer and Manipulation, Program Control.

UNIT IV:

Memory Organization: Memory Hierarchy, Main Memory, Auxiliary memory, Associate Memory, Cache Memory, Cache memories performance considerations, Virtual memories Introduction to Shift registers and RAID

UNIT V:

Input–Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupts, DMA, Input Output Processor, Serial Communication.

Text Books:

1. Digital Logic and Computer Design,Moriss Mano, 11th Edition, Pearson Education.
2. Computer Organization,5thed.,Hamacher,Vranesicand Zaky,TMH,2002
3. Computer System Architecture, 3/e, Moris Mano, Pearson/PHI.

Reference Books:

1. Computer System Organization &Architecture, John D. Carpinelli, Pearson, 2008
2. Computer System Organization, NareshJotwani, TMH, 2009
3. Computer Organization &Architecture: Designing for Performance, 7thed., William Stallings, PHI, 2006
4. Structured Computer Organization – Andrew S. Tanenbaum, 4th Edition, PHI/Pearson.

Web Resources:

1. https://nptelvideos.com/course.php?id=396
2. https://onlinecourses.nptel.ac.in/noc20_cs64/preview
3. https://www.learncomputerscienceonline.com/computer-organization-and- architecture/
4. http://williamstallings.com/COA/COA8e-student/index.html

II Semester	L	T	P	C
	3	0	0	3
24BC2T05 - DATA STRUCTURES				

Course Objectives:

- Solve problems using data structures such as linear lists, stacks, queues, hash tables
- Be familiar with advanced data structures such as binary search trees

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Implement and analyze various search and sorting algorithms, and understand list ADTs including their implementations..	K4
CO2	Implement and apply stack and queue data structures using both arrays and linked lists, and understand their real-world applications.	K4
CO3	Apply hashing techniques and collision resolution strategies to efficiently manage and retrieve data.	K4
CO4	Implement heap operations and apply heaps in problem-solving, and understand the various types of tries.	K3
CO5	Implement and analyze binary trees and binary search trees, and understand basic graph concepts and algorithms for minimum cost spanning trees.	K3

UNIT- I:
Searching: Linear and Binary Search Methods. **Sorting:** Selection Sort, Insertion Sort, Quick Sort, Merge Sort, Heap sort. Introduction to Data Structures: Abstract Data Types (ADTs), The List ADT:Simple Array Implementation of Lists, Simple Linked Lists, Doubly Linked Lists, Circularly Linked Lists.

UNIT-II:
The Stack ADT: The Stack Model, Implementation of Stacks, Applications of Stack.
The Queue ADT: Queue Model, Array Implementation of Queues, Application of Queues, Stacks and Queue. implementation of stacks and queues using linked list.

UNIT- III:
Hashing: Hash Function, Separate Chaining, Collision Resolution-Separate Chaining, Open Addressing: Linear Probing, quadratic probing, Double Hashing, rehashing, Extendible Hashing.

UNIT- IV:
Heap: Structure Property, Heap-Order Property, Basic Heap Operations. Applications: The Selection problem, Event Simulation. Tries, Standard Tries, Compressed Tries, Suffix Tries. (Examples only).

UNIT –V:
Trees: Binary Trees- traversals, Implementation, Expression Trees. Binary Search Trees- find, findMin and find Max, insert, delete operations.
Graphs-Basic Concepts , Storage Structures and Traversals, minimum cost spanning trees

Text Books:

1. Data Structures and Algorithm Analysis, 4th Edition, Mark Allen Weiss, Pearson.
2. Data Structures: A Pseudo Code Approach with C, 2nd Edition, Richard F.Gilberg, &Behrouz A. Forouzon, Cengage.

3. C and Data Structures:A Snap Shot Oriented Treatise Using Live Engineering Examples,N.B.Venkateswarulu,E.V.Prasad, S Chand &Co ,2009.

References Books:

1. Data Structures, Algorithms and Applications in java, 2/e, SartajSahni, University Press.
2. Data Structures using C, 2/e, ReemaThareja

Web Resources:

1. https://archive.nptel.ac.in/courses/106/102/106102064/
2. https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/video_galleries/lecture-videos/
3. https://www.cs.usfca.edu/~galles/visualization/Algorithms.html
4. https://visualgo.net/en
5. https://elearn.daffodilvarsity.edu.bd/course/view.php?id=11771

II Semester	L	T	P	C
	3	0	0	3
24BC2T06 - OOPS THROUGH C++				

Course Objectives:
This course is designed to provide a comprehensive study of the C programming language.

- It stresses the strengths of C, which provide students with the means of writing efficient, maintainable and portable code.
- The nature of C language is emphasized in the wide variety of examples and applications.
- To know about some popular programming languages and how to choose Programming language for solving a problem.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the evolution and fundamentals of C++ and OOP	K1-K2
CO2	Create classes, objects, and manage constructors/destructors	K3
CO3	Implement operator overloading and inheritance	K3-K4
CO4	Apply pointers and virtual functions for polymorphism	K4-K5
CO5	Utilize templates and handle exceptions effectively	K4-K5

UNIT I:
Introduction to C++: Difference between C and C++, Evolution of C++, The Object-Oriented Technology, Disadvantage of Conventional Programming, Key Concepts of Object-Oriented Programming, Advantage of OOP, Object Oriented Language.

UNIT II:
Classes and Objects &Constructors and Destructor: Classes in C++, Declaring Objects, Access Specifiers and their Scope, Defining Member Function, Overloading Member Function, Nested class, Constructors and Destructors, Introduction, Characteristics of Constructor and Destructor, Application with Constructor, Constructor with Arguments, parameterized Constructor, Destructors, Anonymous Objects.

UNIT III:
Operator Overloading and Type Conversion & Inheritance: The Keyword Operator, Overloading Unary Operator, Operator Return Type, Overloading Assignment Operator (=), Rules for Overloading Operators, Inheritance, Reusability, Types of Inheritance, Diamond Problem, Virtual Base Classes, Object as a Class Member, Abstract Classes, Inheritance, Advantages of Inheritance and Disadvantages of Inheritance.

UNIT IV:
Pointers & Binding Polymorphisms and Virtual Functions: Pointer, Features of Pointers, Pointer Declaration, Pointer to Class, Pointer Object, The this Pointer, Pointer to Derived Classes and Base Class, Binding Polymorphisms and Virtual Functions, Introduction, Binding in C++, Virtual Functions, Rules for Virtual Function, Virtual Destructor.

UNIT V:
Templates, Exception Handling: Generic Programming with Templates, Need for Templates, Definition of class Templates, Normal Function Templates, Over Loading of Template Function, Bubble Sort Using Function Templates, Difference Between Templates and Macros, Linked Lists with Templates, Exception Handling, Principles of Exception Handling, The Keywords try Exceptions. throw and catch, Multiple

Catch Statement, Specifying

Text Books:

1. E. Balaguruswamy, Object Oriented Programming with C++, 6/e, McGraw Hill, 2013.
2. A First Book of C++, Gary Bronson, Cengage Learning.
3. The Complete Reference C++, Herbert Schildt, TMH.
4. Programming in C++, Ashok N Kamthane, Pearson 2nd Edition

Reference Books:

1. Object Oriented Programming with C++ by ReemaThareja, OXFORD University Press
2. Object Oriented Programming C++, Joyce Farrell, Cengage.
3. C++ Programming: from problem analysis to program design, DS Malik, Cengage Learning

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc22_cs103/preview
2. https://ocw.mit.edu/courses/6-088-introduction-to-c-memory-management-and-c- object-oriented-programming-january-iap-2010/pages/lecture-notes/
3. https://see.stanford.edu/Course/CS106B
4. https://www.udemy.com/course/object-oriented-c-plus-plus-programming/
5. https://wiingy.com/learn/cpp/cpp-concepts/

II Semester	L	T	P	C
	0	0	3	1.5
24BC2L01 - DATA STRUCTURES THROUGH C LAB				

Course Objectives:
From the course the student will

- Solve problems using data structures such as linear lists, stacks, queues, hash tables
- Be familiar with advanced data structures such as balanced search trees, AVL Trees, and B Trees.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand and implement basic linked list operations, including singly, doubly, and circular linked lists	K3
CO2	Develop algorithms to reverse, sort, merge, and remove duplicates in linked lists.	K4
CO3	Implement and perform stack and queue operations using linked lists, including infix to postfix conversion and expression evaluation.	K3
CO4	Apply recursive and non-recursive methods to implement Binary Search Trees and AVL Trees.	K4
CO5	Develop and implement sorting algorithms, including Selection, Insertion, Merge, Heap, and Quick Sort	K4

List of experiments:

Experiment 1:
Write a C program to perform various operations on a single linked list.

Experiment 2:
Write a C program for the following:
a) Reverse a linked list.
b) Sort the data in a linked list

Experiment 3:
Write a C program for the following:
a) Remove duplicates from a linked list.
b) Merge two linked lists.

Experiment 4:
Write a C program to perform various operations on a doubly linked list.

Experiment 5:
Write a C program to perform various operations on a circular linked list.

Experiment 6:
Write a C program for performing various operations on a stack using a linked list.

Experiment 7:
Write a C program for performing various operations on a queue using a linked list.

Experiment 8:
Write a C program for the following using a stack:
a) Infix to postfix conversion.

b) Expression evaluation.
c) Obtain the binary number for a given decimal number.

Experiment 9:

Write a C program to implement various operations on a Binary Search Tree using Recursive and Non-Recursive methods.

Experiment 10:

Write a C program to implement Selection Sort & Insertion Sort for given elements.

Experiment 11:

Write a C program to implement Merge & Heap Sort for given elements.

Experiment 12:

Write a C program to implement Quick Sort for given elements.

Experiment 13:

Write a C program to implement various operations on AVL trees.

Experiment 14:

Write a C program to perform the following operations:

a) Insertion into a B-tree.
b) Searching in a B-tree.

Experiment 15:

Write a C program for the implementation of recursive and non-recursive functions for Binary Tree Traversals.

Web Resources:

1. https://ds1-iiith.vlabs.ac.in/
2. https://profile.iiita.ac.in/bibhas.ghoshal/teaching_ds_lab.html
3. https://moodle.sit.ac.in/blog/data-structures-laboratory/
4. https://dsalab.netlify.app/
5. https://www.vtuloop.com/data-structure-lab-programs-all/

II Semester	L	T	P	C
	0	0	3	1.5
24BC2L02 - OOPS THROUGH C++ LAB				

Course Objectives:

- To strengthen their problem solving ability by applying the characteristics of an object- oriented approach.
- To introduce object oriented concepts in C++ and Java

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand and implement basic concepts of C++ such as input/output, variables, control flow, and expressions.	K2
CO2	Apply C++ programming concepts to implement functions, classes, object-oriented principles, and memory management.	K3
CO3	Analyze and implement advanced OOP features like inheritance, polymorphism, operator overloading, and templates.	K4
CO4	Develop solutions to problems using STL (Standard Template Library) and demonstrate exception handling mechanisms in C++	K3
CO5	Demonstrate the ability to use inheritance and polymorphism for code reuse and runtime behavior customization.	K4

Exercise 1: (Basics)
Write a Simple Program on printing “Hello World” and “Hello Name” where name is the input from the user

a) Convert any two programs that are written in C into C++.
b) Write a description of using g++ (150 Words)

Exercise 2: (Expressions Control Flow)

a) Write a Program that computes the simple interest and compound interest payable on principal amount (in Rs.) of loan borrowed by the customer from a bank for a given period of time (in years) at specific rate of interest. Further determine whether the bank will benefit by charging simple interest or compound interest.
b) Write a Program to calculate the fare for the passenger starveling in a bus. When a Passenger enters the bus, the conductor asks “What distance will you travel?” On knowing distance from passenger (as an approximate integer), the conductor mentions the fare to the passenger according to following criteria.

Exercise 3: (Variables, Scope, Allocation)

a) Write a program to implement call by value and call by reference using reference variable.
b) Write a program to illustrate scope resolution, new and delete Operators. (Dynamic Memory Allocation)
c) Write a program to illustrate Storage classes
d) Write a program to illustrate Enumerations

Exercise 4: (Functions)
Write a program illustrating Inline Functions

a) Write a program illustrates function overloading. Write 2 overloading functions for power.
b) Write a program illustrates the use of default arguments for simple interest function.

Exercise 5: (Functions –Exercise Continue)

a) Write a program to illustrate function overloading. Write 2 overloading functions for adding two numbers
b) Write a program illustrate function template for power of a number.
c) Write a program to illustrate function template for swapping of two numbers.

Exercise 6: (Classes Objects)
Create a Distance class with:

- feet and inches as data members
- member function to input distance
- member function to output distance
- member function to add two distance objects

a) Write a main function to create objects of DISTANCE class. Input two distances and output the sum.
b) Write a C++ Program to illustrate the use of Constructors and Destructors (use the above program.)
c) Write a program for illustrating function overloading in adding the distance between objects (use the above problem).
d) Write a C++ program demonstrating a Bank Account with necessary methods and variables

Exercise 7: (Access)
Write a program for illustrating Access Specifiers public, private, protected

a) Write a program implementing Friend Function
b) Write a program to illustrate this pointer
c) Write a Program to illustrate pointer to a class

Exercise 8: (Operator Overloading)

a) Write a program to Overload Unary, and Binary Operators as Member Function, and Non - Member Function.
    i Unary operator as member function
    ii Binary operator as nonmember function
b) Write a C ++ program to implement the overloading assignment = operator

Exercise 9: (Inheritance)

a) Write C++ Programs and incorporating various forms of Inheritance
b) Single Inheritance ii) Hierarchical Inheritance iii) Multiple Inheritances

Exercise 10: (Inheritance –Continued)

a) Write C++ Programs and incorporating various forms of Inheritance

i. Multi-level inheritance ii) Hybrid inheritance

b) Write a program to show Virtual Base Class

Exercise 11: (Inheritance –Continued)

a) Write a Program in C++ to illustrate the order of execution of constructors and destructors in inheritance
b) Write a Program to show how constructors are invoked in derived class.

Exercise 12: (Polymorphism)

a) Write a program to illustrate runtime polymorphism
b) Write a program to illustrate this pointer
c) Write a program illustrates pure virtual function and calculate the area of different shapes by using abstract class.

Exercise 13: (Templates)

a) Write a C++ Program to illustrate template class
b) Write a Program to illustrate class templates with multiple parameters
c) Write a Program to illustrate member function templates

Exercise 14: (Exception Handling)

a) Write a Program for Exception Handling Divide by zero
b) Write a Program to rethrow an Exception

Exercise 15: (STL)

a) Write a Program to implement List and List Operations
b) Write a Program to implement Vector and Vector Operations

Web Resources:

1. https://eecs.wsu.edu/~nroy/courses/cpts122/labs/
2. https://www.coursera.org/learn/c-plus-plus-a
3. https://www.educative.io/courses/learn-object-oriented-programming-in-cpp
4. https://www.udemy.com/course/fundamentals-of-object-oriented-programing- cplusplus/

II Semester	L	T	P	C
	2	0	0	2
24BC2T07 - INDIAN CONSTITUTION				

Course Objectives:

Students will be able to:

- Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective.
- To address the growth of Indian opinion regarding modern Indian intellectuals' constitutional role and entitlement to civil and economic rights as well as the emergence of nationhood in the early years of Indian nationalism.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the historical background and evolution of the Indian Constitution, including its drafting process and philosophical underpinnings.	K1
CO2	Analyze the Constitutional Rights and Duties, including fundamental rights, directive principles, and duties enshrined in the Indian Constitution.	K3
CO3	Comprehend the functioning and significance of the Election Commission and its role in upholding democratic processes, as well as the institutions focused on the welfare of marginalized groups.	K2
CO4	Evaluate the roles, powers, and responsibilities of the different organs of governance, including the Parliament, Executive, and Judiciary.	K4
CO5	Gain insight into the structure and functioning of local governance, including municipalities, panchayati raj, and their importance in grassroots democracy.	K3

UNIT – I:

History of Making of the Indian Constitution: History, Drafting Committee,(Composition & Working), Philosophy of the Indian Constitution: Preamble, Salient, Features.

UNIT – II:

Contours of Constitutional Rights &Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties.

UNIT – III:

Election Commission: Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.

UNIT – IV:

Organs of Governance: Parliament, Composition, Qualifications and Disqualifications, Powers and Functions, Executive: President, Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions.

UNIT – V:

Local Administration: District's Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative CEO of Municipal Corporation. Panchayati raj: Introduction, PRI: ZilaPanchayat. Elected officials and their roles, CEO ZilaPanchayat: Position and role. Block level: Organizational Hierarchy (Different departments), Village level: Role of Elected and Appointed officials, Importance of grass root democracy.

Text Books:

1. The Constitution of India, 1950 (Bare Act), Government Publication
2. Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, Dr. S. N. Busi, 2015

Reference Books:

1. Indian Constitution Law, 7th Edition. M. P. Jain, Lexis Nexis, 2014.
2. Introduction

III Semester	L	T	P	C
	3	0	0	3
24BC3T01 - OPERATING SYSTEM				

Course Objectives:
The main objectives of the course is to make student:

- Understand the basic concepts and principles of operating systems, including process management, memory management, file systems, and Protection.
- Make use of process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.
- Illustrate different conditions for deadlock and their possible solutions.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the fundamental concepts and types of Operating Systems, their structure, services, and the role of system calls.	K1
CO2	Explain and analyze the process management concepts including process states, scheduling algorithms, and inter process communication.	K3
CO3	Demonstrate knowledge of memory management techniques such as paging, virtual memory, and page replacement algorithms.	K2
CO4	Apply synchronization techniques and evaluate solutions to concurrency issues, including deadlock detection and recovery strategies.	K4
CO5	Understand file system structures, access methods, and evaluate file system implementation techniques including allocation and disk scheduling.	K3

UNIT-I:
Introduction to Operating System Concepts: Types of operating systems, operating systems concepts, Evaluation of operating systems, operating systems services, structure of OS, Introduction to System call, System call types.

UNIT-II:
Process Management: Process concept, The process, Process State Diagram, Process control block, Process Scheduling: Scheduling Queues, Schedulers, Operations on Processes, Interprocess Communication, Threading Issues, Scheduling-Basic Concepts, Scheduling Criteria, Scheduling Algorithms.

UNIT-III:
Memory Management: Swapping, Contiguous Memory Allocation, Paging, structure of the Page Table, Virtual Memory Management: Virtual Memory, Demand Paging, Page-Replacement Algorithms.

UNIT-IV:
Concurrency: Process Synchronization, The Critical- Section Problem, Synchronization Hardware, Semaphores, Classic Problems of Synchronization, Monitors, Synchronization examples Principles of deadlock: System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery form Deadlock

UNIT-V:

File System Interface: Concept of a file, Access Methods, Directory structure, File system mounting, file sharing, protection. File System Implementation: File system structure, allocation methods, Disk scheduling.

Text Book:

1. Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin and Greg Gagne 9th Edition, John Wiley and Sons Inc., 2012.

2. Operating Systems – Internals and Design Principles, William Stallings, 7th Edition, Prentice Hall, 2011.

Reference Books:

1. Operating Systems: A Concept Based Approach, D.M Dhamdhere, 3rd Edition, McGraw-Hill, 2013

Online Learning Resources:

1. https://nptel.ac.in/courses/106/106/106106144/
2. http://peterindia.net/OperatingSystems.html

III Semester	L	T	P	C
	3	0	0	3
24BC3T02 - PYTHON PROGRAMMING				

Course Objectives:
The main objectives of the course is to make student

- understand and use scripting Language
- to expose various problems solving approaches of computer science Syllabus

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the basics of Python programming, including its history, features, syntax, keywords, variables, and script execution.	K1
CO2	Apply Python's operators, data types, and control flow mechanisms to solve simple computational problems.	K3
CO3	Use built-in data structures such as lists, tuples, sets, and dictionaries effectively in programs.	K2
CO4	Develop modular programs using functions, including parameter passing, anonymous functions, and variable scope handling.	K4
CO5	Demonstrate proficiency in using modules and packages, and handle errors and exceptions using Python's built-in mechanisms.	K3

UNIT – I:
Introduction: History of Python, Python Language, Features of Python, Applications of Python, Using the REPL(Shell), Running Python Scripts, Variables, Assignment, Keywords, Input-Output, Indentation.

UNIT – II:
Types, Operators and Expressions: Types: Integers, Strings, Booleans; Operators Arithmetic Operators, Comparison (Relational) Operators, Assignment Operators, Logical Operators, Bitwise Operators, Membership Operators, Identity Operators, Expressions and order of evaluations, Control Flow- if, if-elif-else, for, while, break, continue.

UNIT – III:
Data Structures Lists - Operations, Slicing, Methods; Tuples, Sets, Dictionaries, Sequences. Comprehensions.

UNIT – IV:
Functions: Defining Functions, Calling Functions, Passing Arguments, Keyword Arguments, Variable-length arguments, Anonymous Functions, Fruitful Functions (Function Returning Values), Scope of the Variables in a Function - Global and Local Variables.

UNIT – V:
Modules: Creating modules, import statement, from. import statement, name spacing, Python packages, Introduction to PIP, Installing Packages via PIP, Using Python Packages Error and Exceptions: Difference between an error and Exception, Handling Exception, try except block, Raising Exceptions, User Defined Exceptions.

Text Books:

1. Core Python Programming, W.Chun, Pearson.
2. Fundamentals of Python First Programs, Kenneth. A. Lambert, Cengage.

Reference Books:

1. Python Programming: A Modern Approach, Vamsi Kurama, Pearson.
2. Think Python, Allen Downey, Green Tea Press
3. Introduction to Python Programming, Gowrishankar.S, Veena A, CRC Press.

Web-Resources:

1. https://onlinecourses.nptel.ac.in/noc20_cs83/preview
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012664745277808640481_shared/overview

III Semester	L	T	P	C
	3	0	0	3
24BC3T03 - DATABASE MANAGEMENT SYSTEM				

Course Objectives:

The main objectives of the course is to make student:

- Understand core database concepts including applications, architecture, data models, and the evolution of database systems.
- Design databases using ER modeling, capturing entities, relationships, and constraints, including advanced ER features.
- Apply the relational model, enforce integrity constraints, and convert ER models into relational schemas
- Use SQL effectively to query, update, and manage databases, including the use of constraints, triggers, and stored procedures.
- Implement normalization techniques to refine schemas, ensure data integrity, and apply various normal forms.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the fundamental concepts of database systems, including architecture, applications, users, and historical development.	K1
CO2	Design conceptual data models using ER and EER diagrams, identifying entities, relationships, and constraints.	K3
CO3	Apply relational model concepts and integrity constraints, and transform ER diagrams into relational schemas.	K2
CO4	Develop and execute SQL queries, implement constraints, triggers, and stored procedures for effective database manipulation.	K4
CO5	Analyze and apply normalization techniques to improve database schema design and ensure data integrity.	K3

UNIT – I:

Overview of Database System: Database System Applications, Purpose of Database Systems, View of Data, Database Languages, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Data Mining and Informational Retrieval, Specialty Databases, Database Users and Administrators, History of Database Systems. [**Text Book -2]**

UNIT –II:

Introduction to Database Design: Database Design and ER Diagrams, Entities, Attributes and Entity Sets, Relationships and Relationship Sets, Additional Features of the ER Model, Conceptual Design with the ER Model, Extended ER features **[Text Book -1]**

UNIT –III:

Relational Model: Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity Constraints, Querying Relational Data, Logical Database Design: ER to Relational, Introduction to Views, Destroying/Altering Tables and Views **[Text Book -1]**

UNIT –IV:

SQL: Queries, Constraints, Triggers: The Form of a Basic SQL Query, UNION, INTERSECT and EXCEPT, Nested Queries, Aggregate Operators, Null Values, Complex Integrity Constraints in SQL, Triggers, Exceptions, Procedures, Functions **[Text Book -1]**

UNIT –V:

Normal Forms: Introduction to Schema Refinement, Functional Dependencies, Reasoning about FDs, Normal Forms, Properties of Decompositions, Normalization. [**Text Book -1]**

Text Books:

1. Data base Management Systems, 3/e, Raghurama Krishnan, Johannes Gehrke, Mc Graw-Hill

2. Data base System Concepts, 6/e, Abraham Silberschatz, Henry F. Korth, S. Sudarshan, Mc Graw-Hill

Reference Books:

1. Database Systems, 6/e Ramez Elmasri, Shamkant B. Navathe, Pearson
2. Introduction to Database Systems, 8/e, C J Date, Pearson
3. Database Systems, 9/e, Carlos Coronel, Steven Morris, Peter Rob, Cengage

Web-Resources:

1. https://nptel.ac.in/courses/106/105/106105175/
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

III Semester	L	T	P	C
	3	0	0	3
24BC3T04 - ARTIFICIAL INTELLIGENCE				

Pre-requisite:

1. Knowledge in Computer Programming
2. A course on "Mathematical Foundations of Computer Science".
3. Background in linear algebra, data structures, algorithms and probability.

Course Objectives:

1. The student should be made to study the concepts of Artificial Intelligence, learn the methods of solving problems using Artificial Intelligence.
2. The student should be made to introduce the concepts of applications of AI, namely game playing, theorem proving, and machine learning.
3. To learn different knowledge representation techniques

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the fundamental concepts of Artificial Intelligence, types of agents, and problem formulation techniques.	K1
CO2	Apply various search strategies, including uninformed, heuristic, and adversarial search algorithms for problem solving and game playing.	K3
CO3	Represent knowledge using logic-based and rule-based systems, and apply probabilistic reasoning under uncertainty.	K2
CO4	Demonstrate understanding of first-order logic and perform inference techniques like unification, forward, and backward chaining.	K4
CO5	Analyze and apply learning strategies such as decision trees, explanation-based learning, statistical methods, and reinforcement learning.	K3

UNIT – I:
Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT – II:
Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Problem reduction, Game Playing- Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.

UNIT – III:
Knowledge Representation: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing

knowledge using rules, rules based deduction systems. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and dempstershafer theory.

UNIT – IV:
Mathematical Logic: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts forward chaining,

UNIT – V:
Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning.

Textbooks:

1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", SecondEdition, Pearson Education.

2. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Mc Graw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel, "Computational Intelligence: a logical approach", Oxford University Press.
2. G. Luger, "Artificial Intelligence: Structures and Strategies for complex problem solving", Fourth Edition, Pearson Education.
3. J. Nilsson, "Artificial Intelligence: A new Synthesis", Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, CENGAGE Learning.

Online Learning Resources:

1. https://ai.google/
2. https://swayam.gov.in/nd1_noc19_me71/preview

III Semester	L	T	P	C
	0	0	3	1.5
24BC3L01 - OPERATING SYSTEMS LAB				

Course Objectives:

To implement the scheduling algorithms, page replacement algorithms, file allocation methods, understand and implement IPC mechanism using named and unnamed pipes.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Implement CPU scheduling algorithms such as FCFS, SJF, Priority, and Round Robin using C programming.	K3
CO2	Simulate different file organization and allocation techniques in C.	K3
CO3	Apply system calls to perform file operations like copying files in a UNIX environment.	K4
CO4	Develop solutions to avoid and prevent deadlocks using Banker's Algorithm.	K4
CO5	Simulate page replacement algorithms such as FIFO, LRU, and LFU for memory management.	K3

List of Experiments:

1. Write C programs to simulate the following CPU scheduling algorithms:
   a) Round Robin b) SJF
2. Write C programs to simulate the following CPU scheduling algorithms:
   a) FCFS b)Priority
3. Write C programs to simulate the following File organization techniques:
   a) Single level directory b) Two level c) Hierarchical
4. Write C programs to simulate the following File allocation methods:
   a) Contiguous b) Linked c) Indexed
5. Write a C program to copy the contents of one file to another using system calls.
6. Write a C program to simulate Bankers Algorithm for Dead Lock Avoidance
7. Write a C program to simulate Bankers Algorithm for Dead Lock Prevention
8. Write C programs to simulate the following page replacement algorithms:
   a) FIFO b) LRU c) LFU

III Semester	L	T	P	C
	0	0	3	1.5
24BC3L02 - PYTHON PROGRAMMING LAB				

Course Objectives:

- To be able to introduce core programming basics and various Operators of Python programming language.
- To demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries
- To understand about Functions, Modules and Regular Expressions in Python Programming

Note : Students are encouraged to practice using Jupyter Notebook and Google Colab.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Demonstrate the use of Python basic data types, control structures, string operations, and date-time manipulation.	K3
CO2	Apply modular programming concepts using functions, recursion, and classes in Python.	K4
CO3	Develop Python programs to perform file operations and data analysis using dictionaries and string processing techniques.	K4
CO4	Analyze and implement mathematical and statistical operations using lists, including mean, median, mode, and matrix operations.	K4
CO5	Implement programs to solve real-time problems using object-oriented and structured programming concepts in Python.	K3

List of Experiments:

Exercise- 1:

a) Write a program to demonstrate different number data types in Python.
b) Write a program to perform different Arithmetic Operations on numbers in Python.
c) Write a program to create, concatenate and print a string and accessing sub-string from a given string.

Exercise- 2:

a) Write a python script to print the current date in the following format “WED 09 02:26:23 IST 2020”
b) Write a Python program to convert temperatures to and from Celsius, Fahrenheit [Formula: c/5 = f-32/9]
c) Write a Python script that prints prime numbers less than 20.

Exercise- 3:

a) Write a python program to find factorial of a number using Recursion.
b) Write a python program to define a module to find Fibonacci Numbers and import the module to another program.
c) Write a Python class to convert an integer to a roman numeral.

Exercise- 4:

a) Write a script named copyfile.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file.
b) Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order.

Exercise- 5:

a) Write a Python class to implement pow(x, n)
b) Write a Python class to reverse a string word by word.
c) Write a Python program to print the following triangle

```
1 2 3 4
  2 3 4
    3 4
      4
```

Exercise- 6:

a) Write a program to count the occurrences of each character in the string and store them in a dictionary data structure
b) Write a program to use split and join methods in the string and trace a birthday with a dictionary data structure.

Exercise- 7 :

a) Write a program to count frequency of characters in a given file. Can you use character frequency to tell whether the given file is a Python program file, C program file or a text file?
b) Write a function nearly equal to test whether two strings are nearly equal. Two strings a and b are nearly equal when a can be generated by a single mutation on b.

Exercise- 8:

a) Find mean, median, mode for the given set of numbers in a list
b) Write a function dups to find all duplicates in the list.
c) Write a function unique to find all the unique elements of a list.

Exercise- 9:

a) Write a program that defines a matrix and prints
b) Write a program to perform addition of two square matrices
c) Write a program to perform multiplication of two square matrices.

III Semester	L	T	P	C
	0	0	3	1.5
24BC3L03 - DATABASE MANAGEMENT SYSTEMS LAB				

Course Objectives:
This Course will enable students to

- Populate and query a database using SQL DDL/DML Commands
- Declare and enforce integrity constraints on a database
- Writing Queries using advanced concepts of SQL
- Programming PL/SQL including procedures, functions, cursors and Triggers

Experiments covering the topics:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions,
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers,

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Design and manipulate relational database schemas using DDL, DML, and various SQL queries including nested and aggregate functions.	K3
CO2	Apply advanced SQL features such as views, constraints, conversion and date functions for efficient database querying.	K4
CO3	Develop PL/SQL programs using control structures, exception handling, cursors, and transactions.	K4
CO4	Implement procedures, functions, and packages for modular and reusable database programming.	K4
CO5	Design and test triggers to handle automatic execution of actions in response to specific database events.	K4

Sample Experiments:

1. Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables), examples using SELECT command.
2. Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example: Select the roll number and name of the student who secured fourth rank in the class.
3. Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.
    i. Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)

ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.

6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.

7. Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE-APPLICATION ERROR.
8. Program development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
9. Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.
10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers

Text Books/Suggested Reading:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
3. Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007

Sample Lab Exercise1: ER Diagram and Relational Schema Design

Objective:
To model a real-world application using ER diagrams and convert it into a relational schema.

Problem Statement:
Design an ER diagram for a Student-Course Enrollment System with the following requirements:

- A student has an ID, name, and department.
- A course has a course ID, name, and credits.
- An instructor teaches multiple courses.
- Students enroll in multiple courses.
- Keep track of which instructor teaches which course.

Instructions:
Draw an ER diagram capturing entities, attributes, and relationships.
Convert the ER diagram into relational schemas with primary and foreign keys.
Implement the schema in SQL.

III Semester	L	T	P	C
	0	0	3	1.5
24BC3L04 - ANIMATION DESIGN				

Course Objectives:
Animation design courses aim to equip students with the skills to create moving images, tell stories through visuals, and work with various animation techniques and software.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Design professional visual elements like logos, brochures, and visiting cards using image editing tools.	K3
CO2	Create and animate graphics using tools like Adobe Flash and apply special effects and filters.	K4
CO3	Model characters and objects (e.g., cars, bikes) and design complex scenes using advanced animation techniques.	K4
CO4	Apply lighting and rendering techniques for realistic indoor and outdoor scenes and synchronize video with audio.	K4
CO5	Develop and integrate special effects like snow, rain, and explosions to enhance multimedia presentations.	K4

List of Experiments:

Experiment 1:
1. Design a logo, brochure, cover letter, visiting cards.
2. Prepare a cutout of some images using Photoshop
3. Place nice background for those images.

Experiment 2:
1. Prepare nice background using gradient tool.
2. Creating Web Banners in Adobe Flash
3. Creating a Logo Animation in Adobe Flash

Experiment 3:
1. Draw Cartoon Animation using reference.
2. Create Lip Sink to Characters
3. Using filters & Special effects

Experiment 4:
1. Create a scene by using Mask layers animation
2. Create any Model of Cars or Bike,
3. Create any model of the male or female character.

Experiment 5:
1. Create a natural outdoor or indoor scene.
2. Render a frame and video of indoor and outdoor scenes.
3. Advance lighting using mental ray render.

Experiment 6:

1. Animate day and night scene of a street with the help of lighting.
2. Title Graphics
3. Video – Audio synchronization
4. 30 Second Commercial AD

Experiment 7:

1. Creating an effect of snow or rain or smoke or water.
2. Creating an effect of bomb/explosion.

Experiment 8:

1. Create a natural outdoor or indoor scene.
2. Set light for Day, Night and Morning
3. Render a frame and video of indoor and outdoor scenes.

III Semester	L	T	P	C
	0	1	2	2
24BC3Z01 - DESIGN THINKING AND INNOVATION				

Course Objectives:

- Build mindsets & foundations essential for designers
- Learn about the Human-Centered Design methodology and understand their real-world applications
- Use Design Thinking for problem solving methodology for investigating ill-defined problems.
- Undergo several design challenges and work towards the final design Challenge

Apply Design Thinking on the following Streams to

- Project Stream 1: Electronics, Robotics, IOT and Sensors
- Project Stream 2: Computer Science and IT Applications
- Project Stream 3: Mechanical and Electrical tools
- Project Stream4: Eco-friendly solutions for waste management, infrastructure, safety, alternative energy sources, Agriculture, Environmental science and other fields of engineering.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the core principles of design thinking and develop a designer's mindset to approach real-world problems creatively.	K2
CO2	Apply empathy-driven research by observing, interviewing, and identifying user needs to frame meaningful problem statements.	K3
CO3	Generate innovative ideas and solutions through structured brainstorming and ideation sessions within teams.	K4
CO4	Build and test functional prototypes, gather feedback, and iteratively improve solutions based on user interaction.	K4
CO5	Communicate solutions effectively through visual storytelling, presentations, and documentation, including business model generation.	K3

HOW TO PURSUE THE PROJECT WORK?

- The first part will be learning-based-masking students to embrace the methodology by exploring all the phases of design thinking through the wallet/ bag challenge and podcasts.
- The second part will be more discussion-based and will focus on building some necessary skills as designers and learning about complementary material for human-centered design.
- The class will then divide into teams and they will be working with one another for about 2 – 3 weeks. These teams and design challenges will be the basis for the final project and final presentation to be presented.
- The teams start with Design Challenge and go through all the phases more in depth from coming up with the right question to empathizing to ideating to prototyping and to testing.

- Outside of class, students will also be gathering the requirements, identifying the challenges, usability, importance etc
- At the end, Students are required to submit the final reports, and will be evaluated by the faculty.

TASKS TO BE DONE:

Task 1: Everyone is a Designer

- Understand class objectives & harness the designer mindset

Task 2: The Wallet/Bag Challenge and Podcast

- Gain a quick introduction to the design thinking methodology.
- Go through all stages of the methodology through a simple design Challenge.
- Podcast: Observe, Listen and Engage with the surrounding environment and identify a design challenge.

Task 3: Teams & Problems (Brain Storming)

- Start Design Challenge and learn about teams & problems through this
- Foster team collaboration, find inspiration from the environment and learn how to identify problems

Task 4: Empathizing (Brain Storming)

- Continue Design Challenge and learn empathy
- Learn techniques on how to empathize with users
- Go to the field and interview people in their environments
- Submit Activity Card

Task 5: Ideating (Business Planning)

- Continue Design Challenge and learn how to brainstorm effectively
- Encourage exploration and foster spaces for brainstorming
- Submit Activity Card

Task 6: Prototyping (Business Model generation)

- Continue Design Challenge and learn how to create effective prototypes
- Build tangible models and use them as communication tools
- Start giving constructive feedback to classmates and teammates
- Submit Activity Card

Task 7: Testing

- Finish Design Challenge and iterate prototypes and ideas through user feedback
- Evolve ideas and prototypes through user feedback and constructive criticism
- Get peer feedback on individual and group performance
- Submit Activity Card

Task 8: Pitching

- Prepare poster to communicate your idea, how it works, why it counts, and who it benefits.
- Final Report Submission and Presentation

Task 9: Case Study

- Make students groups. Each group will generate a business model for unsolved problems in their organization and present.

Note: The colleges may arrange for Guest Speakers from Various Design Fields: Graphic Design, Industrial Design, Architecture, Product Design, Organizational Design, etc to enrich the students with Design Thinking Concept.

References:

1. Tom Kelly, The Art of Innovation: Lessons in Creativity From IDEO, America's Leading Design Firm (Profile Books, 2002)
2. Tim Brown, Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation (HarperBusiness, 2009)
3. Jeanne Liedtka, Randy Salzman, and Daisy Azer, Design Thinking for the Greater Good: Innovation in the Social Sector (Columbia Business School Publishing, 2017)

Other Useful Design Thinking Frameworks and Methodologies:

- Human-Centered Design Toolkit (IDEO);
  https://www.ideo.com/post/design-kit
- Design Thinking Boot Camp Bootleg (Stanford D-School);
  https://dschool.stanford.edu/resources/the-bootcamp-bootleg
- Collective Action Toolkit (frogdesign);
  https://www.frogdesign.com/wpcontent/
  - uploads/2016/03/CAT_2.0_English.pdf
- Design Thinking for Educators (IDEO);
  https://designthinkingforeducators.com/

IV Semester	L	T	P	C
	3	0	0	3
24BC4T01 - COMPUTER NETWORKS				

Course Objectives:

The main objectives of the course is to provide an understanding of computer networking theory, including principles embodied in the protocols designed for the application layer, transport layer, network layer, and data link layer of a networking stack

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the basics of data communication, network types, models like OSI and TCP/IP, and physical layer performance metrics.	K2
CO2	Analyze data link layer functions including error detection/correction techniques and media access control protocols.	K3
CO3	Describe and evaluate network layer protocols, IP addressing, and routing algorithms including IPv4 and IPv6.	K3
CO4	Explain transport layer protocols (UDP/TCP) and their features such as flow control, congestion control, and error control.	K2
CO5	Explore application layer protocols and services including DNS, WWW, Email, and techniques to improve Quality of Service (QoS).	K3

UNIT – I:

Introduction: Data Communications, Networks, Network Types, Internet History, Network Models: Protocol Layering, The OSI Model, TCP/IP Protocol Suite, Introduction to Physical Layer: Transmission Impairments, Data Rate Limits, Performance.

UNIT – II:

Introduction to Data-Link- Layer: Link-Layer Addressing, Error Detection and Correction: Block Coding, Cyclic Codes, Checksum Data Link Control: Data- Link Layer Protocols, HDLC, Point-To-Point (PPP), Media Access Control (MAC): ALOHA, CSMA, CSMA/CD, CSMA/CA, Reservation, Polling, Token Passing, FDMA, TDMA, CDMA

UNIT – III:

Introduction to Network Layer: Network-Layer Services, Packet Switching, Network-Layer Performance, IPV4 Addresses, Network Layer Protocols: Internet Protocol (IP), ICMPv4, Mobile IP, Unicast Routing: Routing Algorithms, Unicast Routing Protocols, Next Generation IP: IPv6 Addressing

UNIT – IV:

Introduction to Transport Layer: Introduction, Transport-Layer Protocols, Transport-Layer Protocols: User Datagram Protocol, Transmission Control Protocol: TCP Services, TCP Features, Segment, A TCP Connection, TCP Congestion Control, Flow Control, Error Control

UNIT – V:

Application Layer: WWW, E-MAIL, Domain Name System (DNS), Quality of Service: Flow Control To Improves QoS, Integrated Services

Text Books:

1. Behrouz A. Forouzan, "Data Communications and Networking", 5th Edition, McGraw HillEducation, 2013.

Reference Books:

1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks", 5th Edition, Prentice Hall,2011.
2. Larry L. Peterson and Bruce S. Davie, "Computer Networks A System Approach", 5th Edition,MKP, 2012.
3. James F. Kurose , Keith W. Ross, " Computer Networking, A Top-Down Approach", 5thEdition, Pearson, 2012.

Web Resources:

1. https://www.geeksforgeeks.org/computer-network-tutorials/
2. https://codescracker.com/networking/

IV Semester	L	T	P	C
	3	0	0	3
24BC4T02 - JAVA PROGRAMMING				

Course Objectives:

The learning objectives of this course are to:

- identify Java language components and how they work together in applications
- learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.
- learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications
- understand how to design applications with threads in Java
- understand how to use Java APIs for program development

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the basic concepts of object-oriented programming, data types, operators, and control structures in Java.	K2
CO2	Apply the concepts of classes, objects, constructors, methods, and method overloading to develop Java programs.	K3
CO3	Implement arrays, inheritance, and interfaces to build modular and reusable Java applications.	K3
CO4	Utilize Java packages, exception handling, and file I/O to develop robust and secure applications.	K4
CO5	Develop multithreaded applications, perform string handling, and interact with databases using JDBC.	K4

UNIT I:

Object Oriented Programming: Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.

Data Types, Variables, and Operators :Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, **Introduction to Operators**, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators.

Control Statements: Introduction, if Expression, Nested if Expressions, if– else Expressions, Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do–while Loop, for Loop, Nested for Loop, For– Each for Loop, Break Statement, Continue Statement.

UNIT II:

Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access
Control, Recursive Methods, Nesting of Methods, Overriding Methods, Attributes Final and Static.

UNIT III: Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three dimensional Arrays, Arrays as Vectors.
Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.
Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.

UNIT IV:
Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes, Auto-boxing and Auto-unboxing, Java util Classes and Interfaces, Formatter Class, Random Class, Time Package, Class Instant (java.time.Instant), Formatting for Date/Time in Java, Temporal Adjusters Class, Temporal Adjusters Class.
Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions. Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java(Text Book 2)

UNIT V:
String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer.
Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi-core Processor, Thread Class, Main Thread-Creation of New Threads, Thread States, Thread Priority- Synchronization, Deadlock and Race Situations, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads.
Java Database Connectivity: Introduction, JDBC Architecture, Installing MySQL and MySQL Connector/J, JDBC Environment Setup, Establishing JDBC Database Connections, ResultSet Interface

Text Books:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. 2Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

References Books:

1. The complete Reference Java, 11thedition, Herbert Schildt,TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Resources:

1. https://nptel.ac.in/courses/106/105/106105191/
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01288 0464547618816347_shared/overview

IV Semester	L	T	P	C
	3	0	0	3
24BC4T03 - WEB TECHNOLOGIES				

Course Objectives:

To Learn PHP language for server side scripting

- To introduce XML and processing of XML Data with Java
- To introduce Server side programming with Java Servlets and JSP
- To introduce Client side scripting with JavaScript.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the fundamentals of the Internet, HTML structure, multimedia elements, forms, and apply CSS for webpage styling.	K2
CO2	Develop structured XML documents with DTDs or Schemas, and parse XML using DOM and SAX parsers in Java.	K3
CO3	Build dynamic web applications using Servlets by handling HTTP requests, sessions, cookies, and database connectivity with JDBC.	K4
CO4	Create interactive web pages using JSP, JavaScript, form validation techniques, and client-side scripting with event handling.	K4
CO5	Develop server-side applications using PHP including form handling, file operations, session and cookie management, and MySQL integration	K4

Unit I:
Web Basics- Introduction, Concept of Internet- History of Internet, Protocols of Internet, World Wide Web, URL, Web Server, Web Browser. **HTML- Introduction**, History of HTML, Structure of HTML Document: Text Basics, Structure of HTML Document: Images and Multimedia, Links and webs, Document Layout, Creating Forms, Frames and Tables, Cascading style sheets.

Unit II:
XML Introduction- Introduction of XMLXML: Introduction to XML, Defining XML tags, their attributes and values, Document Type Definition, XML Schemes, Document Object Model, XHTML Parsing XML Data – DOM and SAX Parsers in java.

Unit III:
Introduction to Servlets: Common Gateway Interface (CGI), Life cycle of a Servlet, deploying a Servlet, The Servlet API, Reading Servlet parameters, Reading Initialization parameters, Handling Http Request & Responses, Using Cookies and Sessions, connecting to a database using JDBC.

Unit IV:
Introduction to JSP: The Anatomy of a JSP Page, JSP Processing, Declarations, Directives, Expressions, Code Snippets, implicit objects, Using Beans in JSP Pages, Using Cookies and session for session tracking, connecting to database in JSP.

Client-side Scripting: Introduction to JavaScript, JavaScript language – declaring variables, scope of variables, functions. event handlers (onClick, on Submit etc.), Document Object Model, Form validation.

Unit V:

Introduction to PHP: Declaring variables, data types, arrays, strings, operators, expressions, control structures, functions, reading data from web form controls like text boxes, radio buttons, lists etc., Handling File Uploads. Connecting to database (MySQL as reference), executing simple queries, handling results, Handling sessions and cookies File Handling in PHP: File operations like opening, closing, reading, writing, appending, deleting etc. on text and binary files, listing directories.

Text Books:

1. Web Technologies, Uttam K Roy, Oxford University Press.
2. The Complete Reference PHP — Steven Holzner, Tata McGraw-Hill.

Reference Books:

1. Web Programming, building internet applications, Chris Bates 2″ edition, Wiley Dreamtech.
2. Java Server Pages —Hans Bergsten, SPD O'Reilly.
3. Java Script, D.Flanagan
4. Beginning Web Programming-Jon Duckett WROX.

IV Semester	L	T	P	C
	3	0	0	3
24BC4T04 - SOFTWARE ENGINEERING				

Course Objectives:
The objectives of this course are to introduce

- Software life cycle models, Software requirements and SRS document
- Project Planning, quality control and ensuring good quality software.
- Software Testing strategies, use of CASE tools, Implementation issues, validation & verification procedures.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Understand the evolution of software engineering and compare various software development life cycle models.	K2
CO2	Apply project estimation techniques, risk management strategies, and formal requirement specification methods.	K3
CO3	Analyze and design function-oriented and agile-based software systems, including user interface design principles.	K4
CO4	Evaluate software through systematic coding practices, testing methods, and quality assurance techniques.	K4
CO5	Understand and analyze software maintenance models and reuse strategies in software development.	K3

UNIT I:
Introduction: Evolution, Software development projects, Exploratory style of software developments, Emergence of software engineering, Notable changes in software development practices, Computer system engineering.
Software Life Cycle Models: Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.

UNIT II:
Software Project Management: Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, risk management.
Requirements Analysis and Specification: Requirements gathering and analysis, Software Requirements Specification (SRS), Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.

UNIT III:
Software Design: Overview of the design process, characterizing a good software design. Layered arrangement of modules, Cohesion and Coupling. approaches to software design.
Agility: Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models, Tool Set for the Agile Process (Text Book 2)
Function-Oriented Software Design: Overview of SA/SD methodology, Structured analysis, Developing the DFD model of a system, Structured design, Detailed design, and Design Review.
User Interface Design: Characteristics of a good user interface, Basic concepts, Types of user interfaces, Fundamentals of component-based GUI development, and user interface design methodology.

UNIT IV:

Coding and Testing: Coding, Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Program analysis tools, Integration testing, Testing object-oriented programs, Smoke testing, and Some general issues associated with testing.

Software Reliability and Quality Management: Software reliability. Statistical testing, Software quality, Software quality management system, ISO 9000. SEI Capability maturity model. Few other important quality standards, and Six Sigma.

UNIT V:

Software Maintenance: Characteristics of software maintenance, Software reverse engineering, Software maintenance process models and Estimation of maintenance cost.

Software Reuse: reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program, A reuse approach, and Reuse atorganization level.

Text Books:

1. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.
2. Software Engineering: A Practitioner's Approach, Roger S. Pressman, 9th Edition, Mc-Graw Hill International Edition.

Reference Books:

1. Software Engineering, Ian Sommerville,10th Edition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

e-Resources:

1. https://nptel.ac.in/courses/106/105/106105182/
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01338269041100390473 5_shared/overview

IV Semester	L	T	P	C
	0	0	3	1.5
24BC4L01 - COMPUTER NETWORKS LAB				

Course Objectives:

Learn basic concepts of computer networking and acquire practical notions of protocols with the emphasis on TCP/IP. It provides a practical approach to Ethernet/Internet networking: networks are assembled, and experiments are made to understand the layered architecture and how do some important protocols work

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Demonstrate the working of networking devices and configure systems to form a LAN.	K3
CO2	Implement and analyze error detection and correction techniques like character stuffing, bit stuffing, checksum, Hamming code, and CRC.	K4
CO3	Apply and simulate flow control protocols such as Stop-and-Wait, Go-Back-N, and Selective Repeat.	K3
CO4	Analyze network congestion control using the Leaky Bucket algorithm.	K4
CO5	Develop and evaluate routing algorithms such as Dijkstra's and Distance Vector Routing.	K5

List of Experiments:

1. Study of Network devices in detail and connect the computers in Local Area Network.
2. Write a Program to implement the data link layer farming methods such as

   (i) Character stuffing ii) bit stuffing.
3. Write a Program to implement data link layer farming method checksum.
4. Write a program for Hamming Code generation for error detection and correction.
5. Write a Program to implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP.
6. Write a Program to implement Sliding window protocol for Go back N.
7. Write a Program to implement Sliding window protocol for Selective repeat.
8. Write a Program to implement Stop and Wait Protocol.
9. Write a program for congestion control using leaky bucket algorithm
10. Write a Program to implement Dijkstra's algorithm to compute the Shortest path through a graph.
11. Write a Program to implement Distance vector routing algorithm by obtaining routing table at each node (Take an example subnet graph with weights indicating delay between nodes).

IV Semester	L	T	P	C
	0	0	3	1.5
24BC4L02 - ANDROID APP DEVELOPMENT				

Course Objectives:

- Understand how Android applications work, their life cycle, manifest, Intents, and using external resources
- Design and develop useful Android applications with compelling user interfaces by using, extending, and creating your own layouts and Views and using Menus.
- Secure, tune, package, and deploy Android applications
- Use Android's communication APIs for SMS, telephony, network management, and internet resources (HTTP).

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Design and develop Android applications using Activities, UI components (TextView, Button, Spinner, ACTV) and layouts.	K3
CO2	Implement data persistence using internal storage, Shared Preferences, SQLite database, and display data using adapters.	K4
CO3	Apply Intents, Fragments, and Telephony services for activity navigation and communication features (SMS, calls, email).	K4
CO4	Demonstrate the usage of multimedia features such as media playback, audio/video recording in Android apps.	K3
CO5	Develop location-aware and sensor-integrated apps using Location, Notification, Wi-Fi, Bluetooth, and Google Maps services.	K5

List of Experiments:

1. a) Create an android application to display “welcome” Text Message.
   b) Create an android application to display “welcome” Message by using Button.
2. Create an android application to call different activities by using Implicit and Explicit Intents.
3. Create an android application to select item from given list by using Auto Complete Text View (ACTV).
4. Create an android application to display dropdown menu items and pick one item by using Spinner Component.
5. Create an android application to display internal storage data using Array Adapter.
6. Create an android application to display internal storage data in vertical format by using Custom Adapter.
7. Create an android application to display WhatsApp videos in grid view by using Custom Adapter.
8. Create an android application to display webpage by using Web view Component.
9. Create an android application to display different webpages in fragments by using Fragments Component.
10. Create an android application to store the data by using Shared Preferences.
11. Create an android application to demonstrate concept of SQLite Database Storage method.
12. Create an android application to perform different types of operations (Send SMS, Making call and sending email) by using Telephony app.

13. Write an android program to develop Media player application.
14. Write an android program to develop Audio Recording application.
15. Write an android program to develop Video Recording application.
16. Create an android application to get latitude and longitude value by using Location Service.
17. Create an android application to get the notifications on Notification Bar by Using Notification Service.
18. Create an android application to display available Wi-Fi devices and Paired Wi-Fi devices by using Wi-Fi Service.
19. Create an android application to get the Bluetooth devices and list of devices using Bluetooth and Vibrator Service.
20. Create an android application to display current location on Google maps by using Google-Maps Service.

Reference Books:

1. Android Application Development (with Kitkat Support), Black Book, Pradeep Kothari.
2. Beginning Android 4 Application Development, Wei-Meng Lee.
3. Android Application Development for Dummies, Michael Burton

IV Semester	L	T	P	C
	0	0	3	1.5
24BC4L03 - JAVA PROGRAMMING LAB				

Course Objectives:

The aim of this course is to

- Practice object-oriented programming in the Java programming Language
- implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism
- Illustrate inheritance, Exception handling mechanism, JDBC connectivity
- Construct Threads, Event Handling, implement packages

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Apply fundamental Java concepts including data types, control structures, classes, constructors, and method overloading.	K3
CO2	Implement Object-Oriented Programming concepts such as inheritance, interfaces, abstract classes, and polymorphism.	K4
CO3	Develop programs using exception handling and multithreading with synchronization techniques.	K4
CO4	Demonstrate the use of Java Packages, String handling, and file operations.	K3
CO5	Build Java applications with database connectivity using JDBC for CRUD operations.	K5

Sample Experiments:
Exercise – 1:

a) Write a JAVA program to display default value of all primitive data type of JAVA
b) Write a java program that display the roots of a quadratic equation ax2+bx=0. Calculate the discriminate D and basing on value of D, describe the nature of root.

Exercise - 2

a) Write a JAVA program to search for an element in a given list of elements using binary search mechanism.
b) Write a JAVA program to sort for an element in a given list of elements using bubble sort
c) Write a JAVA program using String Buffer to delete, remove character.

Exercise - 3

a) Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method.
b) Write a JAVA program implements method overloading.
c) Write a JAVA program to implement constructor.
d) Write a JAVA program to implement constructor overloading.

Exercise - 4

a) Write a JAVA program to implement Single Inheritance
b) Write a JAVA program to implement multi-level Inheritance

c) Write a JAVA program for abstract class to find areas of different shapes

Exercise - 5

a) Write a JAVA program give example for “super” keyword.
b) Write a JAVA program to implement Interface.
c) Write a JAVA program that implements Runtime polymorphism

Exercise - 6

a) Write a JAVA program that describes exception handling mechanism
b) Write a JAVA program Illustrating Multiple catch clauses
    - Write a JAVA program for creation of Java Built-in Exceptions
    - Write a JAVA program for creation of User Defined Exception

Exercise - 7

a) Write a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds, (Repeat the same by implementing Runnable)
b) Write a program illustrating is Alive and join ()
c) Write a JAVA program that import and use the user defined packages

Exercise – 8

a) Write a java program that connects to a database using JDBC
b) Write a java program to connect to a database using JDBC and insert values into it and delete values from it

IV Semester	L	T	P	C
	0	0	3	1.5
24BC4L04 - WEB TECHNOLOGIES LAB				

Course Objectives:

- To implement the web pages using HTML and apply styles.
- Able to develop a dynamic webpage by the use of Java Script.
- Design to create structure of web page, to store the data in web document, and transport information through web.
- Able to write a well formed / valid XML document.

Course Outcomes (Cos): At the end of the course, student will able to

CO No.	Course Outcome	Knowledge Level (K)
CO1	Develop static web pages using HTML and CSS for basic user interface design.	K3
CO2	Manipulate and validate web data using XML and JavaScript.	K4
CO3	Apply server-side scripting using PHP for dynamic content generation and file handling.	K4
CO4	Design and implement dynamic web applications using JSP and Servlets with session tracking and MVC architecture.	K5
CO5	Integrate backend database connectivity using JDBC to store and retrieve user and product information in a web environment.	K5

Experiment 1:
Develop static pages (using HTML and CSS) of an online book store. The pages should resemble: www.flipkart.com The website should consist the following pages.

a) Home page
b) Registration and user Login
c) User Profile Page
d) Books catalog
e) Shopping Cart
f) Payment By credit card
g) Order Conformation

Experiment 2:
Create and save an XML document on the server, which contains 10 users information. Write a program, which takes User Id as an input and returns the user details by taking the user information from the XML document.

Experiment 3:
Write a PHP script that reads data from one file and write into another file.

Experiment 4:
Write a PHP script to print prime numbers between 1-50.

Experiment 5:
Validate the Registration, user login, user profile and payment by credit card pages using JavaScript.

Experiment 6:
Write a PHP script to: a. Find the length of a string. b. Count no of words in a string. c. Reverse a string. d. Search for a specific string.

Experiment 7:
Install TOMCAT web server. Convert the static web pages of assignments 2n into dynamic web pages using servlets and cookies. Hint: Users information (user id, password, credit card number) would be stored in web.xml. Each user should have a separate Shopping Cart.

Experiment 8:
Redo the previous task using JSP by converting the static web pages of assignments 2 into dynamic web pages. Create a database with user information and books information. The books catalogue should be dynamically loaded from the database. Follow the MVC architecture while doing the website.

Experiment 9:
Install a database (Mysql or Oracle). Create a table which should contain at least the following fields: name, password, email-id, phone number (these should hold the data from the registration form). Practice 'JDBC' connectivity.
Write a java program/servlet/JSP to connect to that database and extract data from the tables and display them. Experiment with various SQL queries. Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page.

Experiment 10:
Write a JSP which does the following job: Insert the details of the 3 or 4 users who register with the web site (week9) by using registration form. Authenticate the user when he submits the login form using the user name and password from the database.

IV Semester	L	T	P	C
	0	0	0	2
24BC4I01 - SUMMER INTERNSHIP AND SEMINAR				